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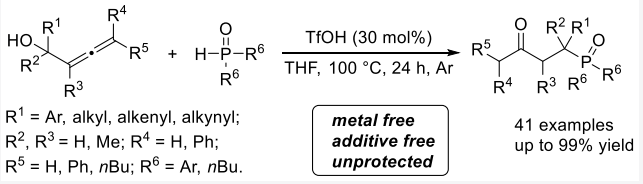
TfOH-Catalyzed Phosphinylation of 2,3-Allenols into γ-Ketophosphine Oxides

Runmin Zhao, Xianhua Huang, Minning Wang, Shanshan Hu, Yuxing Gao,* Pengxiang Xu,* and Yufen Zhao

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ABSTRACT: The first facile and efficient acid-catalyzed direct coupling of a wide range of unprotected 2,3-allenols with arylphosphine oxides was developed, offering a general, one-step approach for the synthesis of structurally diverse γ-ketophosphine oxides accompanied by concurrent C—P/C═O bond formation with remarkable functional group tolerance and complete atom-economy under metal- and additive-free conditions. Mechanistic studies showed that this transformation involved a rearrangement and a phospha-Michael reaction.

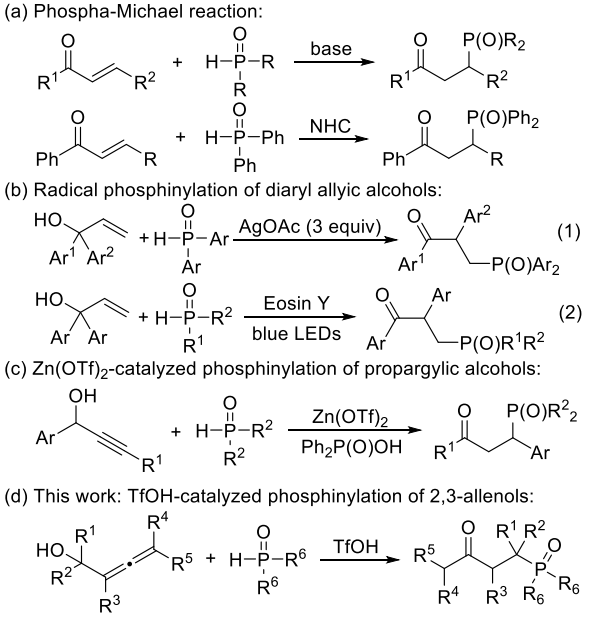


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Organophosphorus compounds have found widespread applications in organic synthesis, medicinal chemistry, agricultural chemistry, and materials science.[1] Among them, γ-ketophosphine oxides have attracted extensive attention in organic and medicinal chemistry over the past few decades due to their unique structural features and diverse intriguing biological properties.[2] Thus, intense research interests have been directed toward developing synthetic approaches to such promising frameworks. The majority of these methods involve the conjugate addition of phosphorus nucleophiles to α,β-unsaturated carbonyl compounds, namely, the phospha-Michael reaction (Scheme 1a), which suffers from the use of a large amount of bases[3] or complex N-heterocyclic carbene[4] as the catalyst and limited substrate scope. To address these disadvantages, a few synthetic methods have emerged in recent years. In 2014, the Ag-promoted radical phosphinylation of α,α-diaryl allylic alcohols with arylphosphine oxides for the preparation of γ-ketophosphine oxides was revealed, which is usually plagued by the use of a large amount of noble Ag salts (Scheme 1b, eq 1).[5] In 2017, our group established a $Zn(OTf)_2$-catalyzed direct coupling of propargylic alcohols with P(O)H compounds to access γ-ketophosphine oxides in the presence of $Ph_2P(O)OH$ (Scheme 1c), but their general use poses severe limitations due to environmentally unfriendly metal catalyst and the limited substrate scope.[6] Very recently, Zhang's group developed Eosin Y-catalyzed phosphinylation of allylic alcohols assisted by blue LEDs (Scheme 1b, eq 2).[7] Although these synthetic methods have proven to be efficient, the development of new catalytic systems with different synthons for the synthesis of diverse γ-ketophosphine oxides for biological screening under metal- and additive-free conditions is still of great significance.

Over the past several years, 2,3-allenols and their derivatives have been employed as structurally diverse and fascinating building blocks for the construction of C—C and C—X bonds,[8] as well as for the preparation of a wide range of valuable molecular frames in modern organic synthesis, such as

Scheme 1. Synthetic Strategies towards γ-Ketophosphine Oxides



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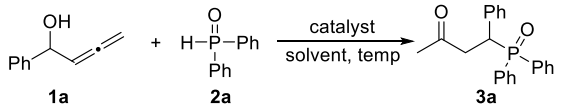
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dihydrofurans,[9] lactams,[10] enynes,[11] 2-pyrrolines,[12] carbazoles,[13] and furanones.[14] Inspired by these precedents and our recent work on C—P bond formation,[15] we disclosed the first example of a single-step and selective synthesis of structurally diverse γ-ketophosphine oxide frameworks via a simple and efficient Brønsted acid-catalyzed direct phosphinylation of unprotected 2,3-allenols under metal-free and additive-free conditions with remarkable functional group tolerance, high regioselectivity, complete atom-economy, and moderate to excellent yields (Scheme 1d). This metal-free strategy provided facile and rapid access to various novel γ-ketophosphine oxides, which were not easily accessible by the known methods, along with the formation of new C—P and C═O bonds.

Our initial investigation focused on the C—P coupling reaction of 1-phenyl-2,3-butadien-1-ol **1a** with diphenylphosphine oxide **2a** as summarized in Table 1. Gratifyingly,

Table 1. Optimization of the Reaction Conditions[a]



1a **2a** **3a**

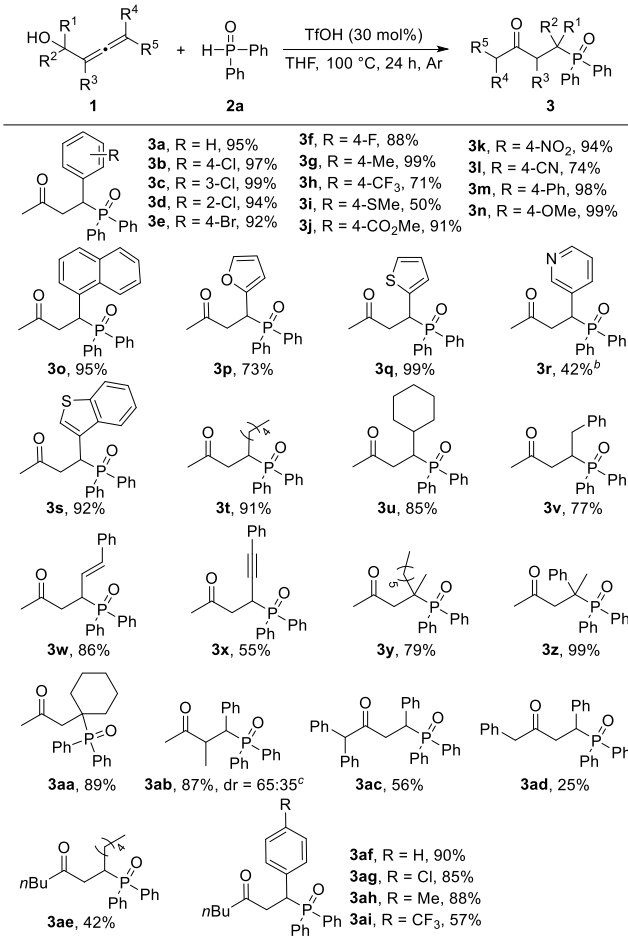
entry	catalyst[b] (mol %)	solvent[b]	temp (°C)	yield (%)
1	$Zn(OTf)_2$ (20)	DCE	100	81
2	$Zn(OTf)_2$(20)	toluene	100	69
3	$Zn(OTf)_2$(20)	dioxane	100	87
4	$Zn(OTf)_2$(20)	THF	100	90
5	$Al(OTf)_3$ (20)	THF	100	88
6	$NHTf_2$(20)	THF	100	76
7	$B(PhF_5)_3$ (20)	THF	100	35
8	$BF_3{\cdot}Et_2O$ (20)	THF	100	64
9	TFA (20)	THF	100	24
10	H_2SO_4 (20)	THF	100	82
11	TfOH (20)	THF	100	92
12		THF	100	0
13	**TfOH (30)**	**THF**	**100**	**95**
14	TfOH (40)	THF	100	93
15	TfOH (10)	toluene	100	58
16	TfOH (10)	DCM	100	27
17	TfOH (10)	DCE	100	89
18	TfOH (30)	THF	90	91
19	TfOH (30)	THF	110	88

[a]Reaction conditions: **1a** (0.4 mmol), **2a** (0.48 mmol), catalyst (mol %), and solvent (2 mL) at the indicated temperature for 24 h under argon. [b]TFA, trifluoroacetic acid; TfOH, trifluoromethanesulfonic acid; OTf, trifluoromethanesulfonate; DCE, dichloroethane; THF, tetrahydrofuran; DCM, dichloromethane.

preliminary attempts using 20 mol % of $Zn(OTf)_2$ as the catalyst in 1,2-dichloroethane (DCE) at 100 °C for 24 h under an argon atmosphere resulted in clean formation of anticipated product **3a** in a high yield of 81% (Table 1, entry 1). Encouraged by this promising result, we turned to screen the effect of other solvents such as toluene, dioxane, and tetrahydrofuran (THF) and found that THF was the optimal solvent for this reaction and could enhance the yield up to 90% (Table 1, entries 1–4). Subsequently, the performance of a series of catalysts was examined. Substituting $Zn(OTf)_2$ for $Al(OTf)_3$, $NHTf_2$, $B(PhF_5)_3$, or $BF_3{\cdot}Et_2O$ as Lewis acid gave rise to a decrease of the yield of **3a** (entries 5–8). A screening of various Brønsted acids including TEA, H_2SO_4, and TfOH as the catalyst was then carried out, with the finding that the formation of **3a** was observed in all cases and simple TfOH gave the best result for this reaction, providing the desired product **3a** in an excellent yield of 92% (entries 9–11). Conversely, no product was detected in the absence of TfOH, and only starting materials were recovered (entry 12). To our delight, increasing the loading of TfOH to 30 mol % could enhance the yield of **3a** up to 95% (entry 13). Yet, further increasing the TfOH loading led to a slight decrease in yield (entry 14). Considering that THF may form a complex with the TfOH and weaken the acidity of the TfOH, we retested the effect of various solvents such as toluene, DCM, and DCE by using 10 mol % TfOH, but none of these gave better results (entries 15–17), indicating that THF was the optimal solvent for the reaction. Finally, the effect of temperature was evaluated, and we found that increasing or decreasing the temperature from 100 °C resulted in the yield reduction (entries 18 and 19).

With the optimized conditions in hand (footnote a, Table 2), the substrate scope of the phosphinylation of various structurally and electronically modulated 2,3-allenols with diphenylphosphine oxide **2a** was explored, and the results are summarized in Table 2. In the beginning, 13 derivatives (**1b**–

Table 2. Substrate Scope of 2,3-Allenols[a]



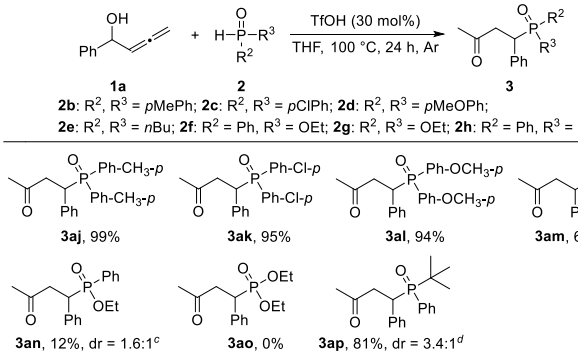
[a]Reaction conditions: **1a** (0.4 mmol), **2a** (0.48 mmol), TfOH (30 mol %), and THF (2 mL) at 100 °C for 24 h under argon. [b]Using 130 mol % of TfOH at 90 °C. [c]Diastereomer ratios were determined by ^{31}P NMR analysis of the crude products.

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1n) of **1a** with different electron-donating or -withdrawing groups on the phenyl ring were tested. Various substituents such as Me, Ph, OMe, SMe, F, Cl, Br, CF_3, CO_2Me, NO_2, and CN were all well tolerated for this method and their electronic nature was not evident for this transformation. The bulky naphthyl-derived 2,3-allenol underwent the coupling to afford the desired product **3o** in 95% yield. In addition, substrates incorporating heterocyclic groups such as furyl (**1p**), thienyl (**1q**), or benzothienyl (**1s**) were also tolerated well. Subjecting pyridyl-substituted 2,3-allenol **1r** to the standard reaction conditions did not afford the targeted product, yet with 1.3 equiv of TfOH at 90 °C, the reaction could give the expected product **3r** in 42% yield. To our delight, aliphatic 2,3-allenols **1t**−**1v** could also successfully be applied to this coupling. Interestingly, **1w** and **1x** having double bond and triple bond at C1 position also worked under the standard conditions. In addition to secondary 2,3-allenols, tertiary 2,3-allenols were also successfully engaged in this reaction, producing the corresponding products **3y**−**3aa** in good to excellent yields of 79−99%. It should be noted that tertiary 2,3-allenols such as 1,1-diphenyl-buta-2,3-dien-1-ol with two phenyls at the C1 position failed to afford the expected product under the optimized conditions, which might be attributed to steric hindrance. After investigating the reactivity of 1-substituted 2,3-allenols, we turned our attention to the use of 1,2-disubstituted 2,3-allenols. When a methyl group was introduced at C2 position, **1ab** could be converted into the desired product **3ab** in high total yield along with a diastereomeric ratio of 65:35. Next, the influence of substituents at the terminal position of the allene moiety was surveyed. 4,4-Diphenyl-substituted 2,3-allenol **1ac** could also participate in the reaction to generate the corresponding product **3ac** in 56% yield. However, when 4-phenyl-substituted 2,3-allenol **1ad** was employed, an unsatisfactory yield of 25% was obtained. It is noteworthy that 2,3-allenols with an alkyl group at their C4 position were also compatible with this protocol and gave the related products **3ae**−**3ai** in moderate to good yields. Fortunately, the product **3e** was recrystallized from acetone as colorless crystals by slow evaporation at room temperature, and the molecular structure of **3e** was confirmed by X-ray diffraction analysis (see the Supporting Information for detailed X-ray crystallography data).[16] The result clearly showed the phosphoryl moiety was preferentially installed at the C1-position of 2,3-allenols in the present coupling reaction.

To gain more insight into the versatility of the reaction, some other P(O)H substrates were also evaluated. As shown in Table 3, apart from **2a**, other diarylphosphine oxides such as **2b**, **2c**, and **2d** were all suitable reaction partners and afforded the desired products **3aj**−**3al** in excellent yields (94%−99%). Moreover, dibutylphosphine oxide **2e** could also be successfully converted into the corresponding product **3am** in 67% yield by increasing the temperature to 140 °C. Nevertheless, ethyl phenylphosphinate **2f** only provided the target product **3an** in a lower yield of 12%. Notably, H-phosphonates like diethyl phosphonate were almost inert in the present catalytic system, probably due to the electronic properties of P(O)H compounds. To examine the stereochemistry of this reaction, optically pure (S)-*tert*-butyl(phenyl)phosphine oxide (99% ee) **2h** was used as the substrate under the standard reaction conditions, but the corresponding product **3ap** was acquired in a racemic form, which was proposed with the low diastereocontrol (3.4:1 dr).

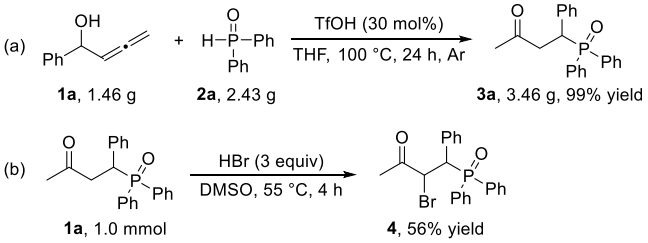
Table 3. Substrate Scope of P(O)H Compounds[a]



[a]Reaction conditions: **1a** (0.4 mmol), **2a** (0.48 mmol), TfOH (30 mol %), and THF (2 mL) at 100 °C for 24 h under argon. [b]At 140 °C. [c]In EtOAc. [d]Diastereomer ratio was determined by ^{1}H NMR analysis of the mixture.

To demonstrate the practical utility of this method, a gram-scale reaction of **1a** (1.46 g, 10 mmol) with **2a** (2.43 g, 12 mmol) was conducted to afford the expected product **3a** (3.46 g) in 99% yield (Scheme 2a), indicating that the reaction could

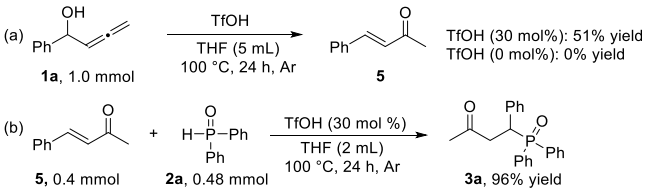
Scheme 2. Application Studies



be effectively scaled up without decreasing the yields. Moreover, the products could be easily transformed to other derivatives.[5,17] For example, **3a** could react with HBr to produce a useful α-bromocarbonyl compound **4** (Scheme 2b).

To gain preliminary insight into the reaction mechanism, several control experiments were performed (Scheme 3).

Scheme 3. Mechanistic Studies

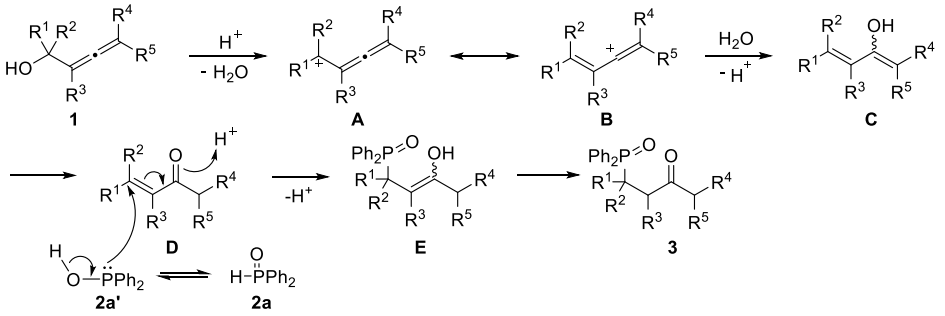


Treatment of **1a** under the standard reaction conditions without **2a**, as expected, produced enone **5** in 51% yield. Yet, no product **5** was observed in the absence of TfOH (Scheme 3a). Further treatment of enone **5** with **2a** could give the targeted product **3a** in 96% yield (Scheme 3b), showing that enone **5** should be the key intermediate in this transformation.

On the basis of the experimental results and related literature reports,[18−20] we proposed a tentative mechanism for this transformation (Scheme 4). Initially, the elimination of the hydroxyl group in the 2,3-allenol moiety easily occurred with the aid of TfOH to produce the carbenium ion **A**, followed by electron transfer of **A** to afford the corresponding

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Scheme 4. Plausible Mechanism



vinyl cation **B**.[18b] Ensuing nucleophilic attack of water led to an enol intermediate **C**, which immediately isomerizes to form the key enone intermediate **D**.[18] Finally, the intermolecular phospha-Michael addition of the phosphorus nucleophile **2′** (tautomerized from **2**[19]) to **D** resulted in the formation of an enol intermediate **E** with the assistance of TfOH,[20] followed by enol−keto tautomerism of **E** to generate the desired product **3**.

In conclusion, we have developed the first facile and efficient TfOH-catalyzed direct phosphinylation of a wide range of 2,3-allenols with H-phosphine oxides, affording a rapid and powerful tool to a structurally diverse array of γ-ketophosphine oxides along with concurrent new C═O and C—P bond formation. Importantly, the present method avoids metals and other additives, and various valuable γ-ketophosphine oxide moieties could be easily obtained through a simple one-step process. Mechanistic investigations showed that this transformation involved a rearrangement to the enone intermediate, followed by the phospha-Michael reaction. In addition, the use of readily available TfOH as the catalyst, operational simplicity, remarkable functional group tolerance, excellent regioselectivity, and complete atom economy mean that this method will open a new avenue to potential biologically active γ-ketophosphine oxide and find widespread applications in synthetic chemistry and medicinal chemistry.

EXPERIMENTAL SECTION

General Information. Unless otherwise stated, all reactions were carried out under dry argon. Substrates **1a**−**1aa**,[21] **1ab**,[12] **1ac**,[22] **1ad**,[23] **1ae**−**1ai**,[24] **2c**−**2d**,[25] **2e**,[26] and **2h**[27] were prepared according to the known reference. Other reagents were obtained from commercial suppliers and used without further purification. All the heating experiments were performed in an oil bath. ^{1}H, ^{13}C, and ^{31}P NMR spectra were recorded on a Bruker AV500 spectrometer using tetramethylsilane (TMS) in $CDCl_3$ as the internal standard. NMR chemical shifts were determined relative to $CDCl_3$ as the internal reference (δ_H = 7.26 ppm, δ_C = 77.16 ppm). NMR data are represented as follows: chemical shift, multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet), coupling constants in Hertz (Hz), integration. HRMS spectra of all products were recorded on a Waters Micromass LCT Premier TOF-MS apparatus. Infrared spectra were recorded on a Nicolet Avatar 330 spectrometer. Melting points were recorded on a MP120 digital melting-point apparatus. The products were purified by column chromatography on silica gel 300−400 mesh. Flash column chromatography purification was carried out by a gradient elution method using ethyl acetate (EA) in light petroleum ether (PE).

*General Procedure for the Synthesis of γ-Ketophosphine Oxides (**3aa**−**3ap**).* To a Schlenk tube equipped with a magnetic stir bar was added 2,3-allenols (0.4 mmol), H-phosphine oxides (0.48 mmol), and TfOH (0.12 mmol) in THF (2 mL) under argon, and the mixture was kept stirring at 100 °C in an oil bath for 24 h. The resulting mixture was cooled down to room temperature and purified by flash chromatography (petroleum ether/ethyl acetate) to afford the corresponding products.

*4-(Diphenylphosphinyl)-4-phenylbutan-2-one (**3a**).*[28] 95% yield, 132 mg; white solid; mp 185.1−187.7 °C. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.95−7.92 (m, 2H), 7.57−7.50 (m, 3H), 7.46−7.43 (m, 2H), 7.33−7.28 (m, 3H), 7.24−7.20 (m, 2H), 7.17−7.10 (m, 3H), 4.23 (ddd, *J* = 10.0, 7.3, 2.7 Hz, 1H), 3.33 (ddd, *J* = 17.6, 10.1, 5.4 Hz, 1H), 2.95 (ddd, *J* = 17.8, 11.2, 2.6 Hz, 1H), 1.94 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 205.3 (d, J_{C-P} = 12.6 Hz), 135.8 (d, J_{C-P} = 5.4 Hz), 132.0 (d, J_{C-P} = 2.7 Hz), 131.8 (d, J_{C-P} = 7.3 Hz), 131.4 (d, J_{C-P} = 2.7 Hz), 131.3 (d, J_{C-P} = 8.2 Hz), 131.0 (d, J_{C-P} = 9.1 Hz), 129.3 (d, J_{C-P} = 103.5 Hz), 129.2 (d, J_{C-P} = 109.9 Hz), 128.3 (d, J_{C-P} = 1.8 Hz), 128.1, 128.0, 127.1 (d, J_{C-P} = 2.7 Hz), 43.5, 41.1 (d, J_{C-P} = 68.3 Hz), 30.6. ^{31}P{^{1}H} NMR ($CDCl_3$, 202 MHz): δ 33.6. IR (film) υ_{max} 2918, 1715, 1202, 1108 cm^{-1}. HRMS (ESI-TOF) *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{21}O_2PH^+$ 349.1352, found 349.1352.

*4-(Diphenylphosphinyl)-4-(4-chlorophenyl)butan-2-one (**3b**).*[29] 97% yield, 149 mg; white solid; mp 215.5−216.5 °C. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.85−7.82 (m, 2H), 7.48−7.37 (m, 5H), 7.27−7.24 (m, 1H), 7.19−7.15 (m, 4H), 7.05−7.03 (m, 2H), 4.13 (ddd, *J* = 10.0, 7.3, 2.6 Hz, 1H), 3.18 (ddd, *J* = 18.0, 10.2, 5.1 Hz, 1H), 2.84 (ddd, *J* = 18.1, 11.0, 2.6 Hz, 1H), 1.86 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 204.9 (d, J_{C-P} = 12.6 Hz), 134.6 (d, J_{C-P} = 5.5 Hz), 133.0 (d, J_{C-P} = 2.8 Hz), 132.1 (d, J_{C-P} = 2.7 Hz), 131.6 (d, J_{C-P} = 2.8 Hz), 131.5, 131.2(d, J_{C-P} = 8.7 Hz), 131.0 (d, J_{C-P} = 5.6 Hz), 130.8 (d, J_{C-P} = 8.6 Hz), 130.7, 128.6 (d, J_{C-P} = 89.2 Hz), 128.6 (d, J_{C-P} = 89.8 Hz), 128.5 (d, J_{C-P} = 1.6 Hz), 43.5, 40.1 (d, J_{C-P} = 68.3 Hz), 30.5. ^{31}P{^{1}H} NMR ($CDCl_3$, 202 MHz): δ 33.2. IR (film) υ_{max} 3053, 2921, 1715, 1176, 1119 cm^{-1}. HRMS (ESI-TOF) *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{20}ClO_2PH^+$ 383.0962, found 383.0962.

*4-(Diphenylphosphinyl)-4-(3-chlorophenyl)butan-2-one (**3c**).* 99% yield, 152 mg; white solid; mp 196.4−198.3 °C. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.86−7.82 (m, 2H), 7.50−7.43 (m, 3H), 7.40−7.37 (m, 2H), 7.28−7.25 (m, 1H), 7.21−7.17 (m, 3H), 7.14−7.11 (m, 1H), 7.01−6.98 (m, 2H), 4.13 (ddd, *J* = 10.1, 7.4, 2.8 Hz, 1H), 3.20 (ddd, *J* = 18.1, 10.1, 5.3 Hz, 1H), 2.86 (ddd, *J* = 18.1, 11.2, 2.4 Hz, 1H), 1.88 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 204.8 (d, J_{C-P} = 12.5 Hz), 138.1 (d, J_{C-P} = 5.4 Hz), 134.0 (d, J_{C-P} = 1.8 Hz), 132.2 (d, J_{C-P} = 2.7 Hz), 131.7 (d, J_{C-P} = 2.7 Hz), 131.2 (d, J_{C-P} = 9.0 Hz), 130.9 (d, J_{C-P} = 8.2 Hz), 129.8 (d, J_{C-P} = 6.4 Hz), 129.5 (d, J_{C-P} = 1.7 Hz), 128.6 (d, J_{C-P} = 94.5 Hz), 128.5 (d, J_{C-P} = 93.7 Hz), 127.8 (d, J_{C-P} = 5.2 Hz), 127.3 (d, J_{C-P} = 2.0 Hz), 43.4, 40.5 (d, J_{C-P} = 67.7 Hz), 30.5. ^{31}P{^{1}H} NMR ($CDCl_3$, 202 MHz): δ 33.2. IR (film) υ_{max} 3056, 2923, 1720, 1179, 1119 cm^{-1}. HRMS (ESI-TOF) *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{20}ClO_2PH^+$ 383.0962, found 383.0965.

*4-(Diphenylphosphinyl)-4-(2-chlorophenyl)butan-2-one (**3d**).* 94% yield, 144 mg; white solid; mp 159.9−161.0 °C. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.94−7.90 (m, 2H), 7.70−7.68 (m, 1H), 7.53−7.47 (m, 3H), 7.35−7.31 (m, 2H), 7.25−7.22 (m, 1H), 7.17−7.14 (m, 1H), 7.13−7.09 (m, 2H), 7.04−6.97 (m, 2H), 4.82 (ddd, *J* = 10.2, 7.1, 3.3 Hz, 1H), 3.26 (ddd, *J* = 17.4, 10.2, 6.4 Hz, 1H), 2.90 (ddd, *J* = 17.9, 10.2, 3.2 Hz, 1H), 1.87 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 204.6 (d, J_{C-P} = 12.3 Hz), 134.8 (d, J_{C-P} = 7.4 Hz), 134.1 (d, J_{C-P} = 4.7 Hz), 132.2 (d, J_{C-P} = 2.7 Hz), 131.6 (d, J_{C-P}

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= 3.0 Hz), 131.4 (d, J_{C-P} = 8.5 Hz), 130.9 (d, J_{C-P} = 9.8 Hz), 129.9 (d, J_{C-P} = 4.5 Hz), 129.3 (d, J_{C-P} = 1.5 Hz), 128.5 (d, J_{C-P} = 137.4 Hz), 128.4 (d, J_{C-P} = 93.7 Hz), 128.3 (d, J_{C-P} = 2.4 Hz), 127.1 (d, J_{C-P} = 2.0 Hz), 43.7, 36.4 (d, J_{C-P} = 67.3 Hz), 30.2. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.7. IR (film) υ_{max} 3057, 2929, 1715, 1187, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{20}ClO_2PH^+$ 383.0962, found 383.0962.

*4-(Diphenylphosphinyl)-4-(4-bromophenyl)butan-2-one (**3e**).*[30] 92% yield, 157 mg; white solid; mp 225.7−226.6 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.93−7.89 (m, 2H), 7.58−7.51 (m, 3H), 7.49−7.45 (m, 2H), 7.38−7.35 (m, 1H), 7.30−7.25 (m, 4H), 7.20−7.18 (m, 2H), 4.19 (ddd, J = 10.0, 7.2, 2.7 Hz, 1H), 3.27 (ddd, J = 18.1, 10.2, 5.1 Hz, 1H), 2.92 (ddd, J = 18.1, 11.0, 2.7 Hz, 1H), 1.96 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.9 (d, J_{C-P} = 12.6 Hz), 135.1 (d, J_{C-P} = 5.5 Hz), 132.1 (d, J_{C-P} = 12.7 Hz), 131.7 (d, J_{C-P} = 2.7 Hz), 131.6 (d, J_{C-P} = 9.4 Hz), 131.4 (d, J_{C-P} = 1.7 Hz), 131.4 (d, J_{C-P} = 5.3 Hz), 131.2 (d, J_{C-P} = 8.8 Hz), 130.9 (d, J_{C-P} = 8.8 Hz), 130.8 (d, J_{C-P} = 4.6 Hz), 128.7 (d, J_{C-P} = 87.4 Hz), 128.6 (d, J_{C-P} = 87.7 Hz), 121.2 (d, J_{C-P} = 3.2 Hz), 43.5, 40.2 (d, J_{C-P} = 67.9 Hz), 30.5. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.0. IR (film) υ_{max} 3050, 2918, 1713, 1175, 1119 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{20}BrO_2PH^+$ 427.0457, found 427.0457.

*4-(Diphenylphosphinyl)-4-(4-fluorophenyl)butan-2-one (**3f**).*[31] 88% yield, 129 mg; white solid; mp 177.4−178.8 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.94−7.91 (m, 2H), 7.58−7.51 (m, 3H), 7.47−7.43 (m, 2H), 7.36−7.33 (m, 1H), 7.30−7.24 (m, 4H), 6.87−6.84 (m, 2H), 4.21 (ddd, J = 1 Hz, 1H), 3.28 (ddd, J = 17.8, 10.2, 5.1 Hz, 1H), 2.92 (ddd, J = 18.0, 10.9, 2.4 Hz, 1H), 1.96 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.1 (d, J_{C-P} = 12.8 Hz), 161.9 (d, J_{C-F} = 246.1 Hz), 132.1 (d, J_{C-P} = 2.8 Hz), 131.8, 131.5 (d, J_{C-P} = 2.8 Hz), 131.3 (d, J_{C-P} = 5.4 Hz), 131.2 (d, J_{C-P} = 8.8 Hz), 131.2 (d, J_{C-P} = 5.4 Hz), 130.9 (d, J_{C-P} = 9.1 Hz), 128.6 (d, J_{C-P} = 95.2 Hz), 128.5 (d, J_{C-P} = 95.4 Hz), 115.4 (d, J_{C-P} = 1.8 Hz), 115.2 (d, J_{C-P} = 1.8 Hz), 43.7, 40.0 (d, J_{C-P} = 68.6 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.4. IR (film) υ_{max} 3056, 2921, 1715, 1175, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{20}FO_2PH^+$ 367.1258, found 367.1257.

*4-(Diphenylphosphinyl)-4-(4-tolyl)butan-2-one (**3g**).*[31] 99% yield, 144 mg; white solid; mp 195.6−198.3 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.93−7.90 (m, 2H), 7.56−7.44 (m, 5H), 7.35−7.32 (m, 1H), 7.26−7.22 (m, 2H), 7.18−7.16 (m, 2H), 6.97−6.95 (m, 2H), 4.19 (ddd, J = 10.2, 7.6, 2.8 Hz, 1H), 3.29 (ddd, J = 17.5, 10.2, 5.4 Hz, 1H), 2.92 (ddd, J = 17.7, 11.1, 2.5 Hz, 1H), 2.22 (s, 3H), 1.94 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.4 (d, J_{C-P} = 12.5 Hz), 136.7 (d, J_{C-P} = 2.7 Hz), 132.6 (d, J_{C-P} = 5.4 Hz), 132.0 (d, J_{C-P} = 4.1 Hz), 131.9 (d, J_{C-P} = 2.6 Hz), 131.4 (d, J_{C-P} = 2.7 Hz), 131.3 (d, J_{C-P} = 8.4 Hz), 131.3 (d, J_{C-P} = 10.0 Hz), 131.0 (d, J_{C-P} = 8.3 Hz), 129.6 (d, J_{C-P} = 5.6 Hz), 129.1 (d, J_{C-P} = 1.5 Hz), 128.5 (d, J_{C-P} = 97.0 Hz), 128.4 (d, J_{C-P} = 98.0 Hz), 43.6, 40.4 (d, J_{C-P} = 68.7 Hz), 30.6, 21.0. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.4. IR (film) υ_{max} 3056, 2921, 1709, 1174, 1120 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_2PH^+$ 363.1508, found 363.1508.

*4-(Diphenylphosphinyl)-4-(4-(trifluoromethyl)phenyl)butan-2-one (**3h**).* 71% yield, 118 mg; white solid; mp 243.9−245.7 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.95−7.91 (m, 2H), 7.60−7.52 (m, 3H), 7.48−7.40 (m, 6H), 7.37−7.34 (m, 1H), 7.27−7.24 (m, 2H), 4.29 (ddd, J = 10.0, 7.1, 2.8 Hz, 1H), 3.32 (ddd, J = 18.3, 10.2, 5.0 Hz, 1H), 2.97 (ddd, J = 18.3, 11.2, 2.8 Hz, 1H), 1.97 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.7 (d, J_{C-P} = 12.2 Hz), 140.4−140.3 (m), 132.2 (d, J_{C-P} = 2.5 Hz), 131.7 (d, J_{C-P} = 2.6 Hz), 131.4 (d, J_{C-P} = 15.9 Hz), 131.2 (d, J_{C-P} = 8.8 Hz), 130.8 (d, J_{C-P} = 9.0 Hz), 130.6 (d, J_{C-P} = 11.6 Hz), 130.3 (d, J_{C-P} = 5.5 Hz), 129.2, 128.7 (d, J_{C-P} = 93.0 Hz), 128.6 (d, J_{C-P} = 93.6 Hz), 126.3 (q, J_{C-F} = 272.5 Hz), 125.2 (q, J_{C-F} = 1.9 Hz), 43.5, 40.7 (d, J_{C-P} = 67.1 Hz), 30.4. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.0. IR (film) υ_{max} 3050, 2916, 1714, 1176, 1111 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{20}F_3O_2PH^+$ 417.1226, found 417.1229.

*4-(Diphenylphosphinyl)-4-(4-methylthiophenyl)butan-2-one (**3i**).* 50% yield, 79 mg; white solid; mp 220.2−220.6 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.93−7.90 (m, 2H), 7.58−7.51 (m, 3H), 7.49−7.45 (m, 2H), 7.37−7.34 (m, 1H), 7.28−7.21 (m, 4H), 7.05−7.04 (m, 2H), 4.17−4.20 (m, 1H), 3.28 (ddd, J = 17.6, 10.2, 5.2 Hz, 1H), 2.92 (ddd, J = 17.7, 11.0, 2.1 Hz, 1H), 2.40 (s, 3H), 1.95 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.2 (d, J_{C-P} = 12.5 Hz), 137.3, 132.5 (d, J_{C-P} = 5.1 Hz), 132.1, 131.7, 131.6, 131.3 (d, J_{C-P} = 8.1 Hz), 131.0 (d, J_{C-P} = 8.4 Hz), 130.1 (d, J_{C-P} = 5.1 Hz), 128.6 (d, J_{C-P} = 90.9 Hz), 128.5 (d, J_{C-P} = 91.2 Hz), 126.4, 43.6, 40.3 (d, J_{C-P} = 70.2 Hz), 30.6, 15.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.3. IR (film) υ_{max} 2922, 1714, 1172, 1118 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_2PSH^+$ 395.1229, found 395.1229.

*Methyl 4-(1-(Diphenylphosphinyl)-3-oxo-3-methylpropyl)-benzoate (**3j**).* 91% yield, 147 mg; yellow solid; mp 203.2−206.1 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.96−7.92 (m, 2H), 7.84−7.83 (m, 2H), 7.58−7.52 (m, 3H), 7.48−7.45 (m, 2H), 7.40−7.39 (m, 2H), 7.34−7.30 (m, 1H), 7.25−7.23 (m, 2H), 4.32−4.28 (m, 1H), 3.85 (s, 3H), 3.34 (ddd, J = 17.5, 10.0, 5.1 Hz, 1H), 3.00−2.95 (m, 1H), 1.95 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.8 (d, J_{C-P} = 12.5 Hz), 166.8, 141.5 (d, J_{C-P} = 5.3 Hz), 132.2 (d, J_{C-P} = 1.8 Hz), 131.7 (d, J_{C-P} = 2.7 Hz), 131.5 (d, J_{C-P} = 3.5 Hz), 131.2 (d, J_{C-P} = 9.0 Hz), 130.8 (d, J_{C-P} = 9.1 Hz), 130.7, 129.7 (d, J_{C-P} = 5.6 Hz), 129.5, 128.9 (d, J_{C-P} = 2.1 Hz), 128.6 (d, J_{C-P} = 93.8 Hz), 128.5 (d, J_{C-P} = 94.5 Hz), 52.0, 43.4, 41.0 (d, J_{C-P} = 67.0 Hz), 30.5. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.9. IR (film) υ_{max} 3056, 2917, 1719, 1182, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{23}O_4PH^+$ 407.1407, found 407.1406.

*4-(Diphenylphosphinyl)-4-(4-nitrophenyl)butan-2-one (**3k**).*[31] 94% yield, 148 mg; pale yellow solid; mp 247.7−249.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 8.02−8.00 (m, 2H), 7.97−7.93 (m, 2H), 7.60−7.48 (m, 7H), 7.38−7.35 (m, 1H), 7.30−7.26 (m, 2H), 4.36 (ddd, J = 9.7, 6.9, 2.4 Hz, 1H), 3.35. (ddd, J = 18.4, 10.2, 5.0 Hz, 1H), 3.01 (ddd, J = 18.5, 10.9, 2.5 Hz, 1H), 1.99 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.4 (d, J_{C-P} = 12.1 Hz), 146.9 (d, J_{C-P} = 2.1 Hz), 144.1 (d, J_{C-P} = 15.4 Hz), 132.4 (d, J_{C-P} = 2.7 Hz), 131.9 (d, J_{C-P} = 2.5 Hz), 131.1 (d, J_{C-P} = 8.5 Hz), 131.0, 130.7 (d, J_{C-P} = 9.0 Hz), 130.6 (d, J_{C-P} = 5.4 Hz), 130.3 (d, J_{C-P} = 21.4 Hz), 128.8 (d, J_{C-P} = 87.1 Hz), 128.7 (d, J_{C-P} = 86.2 Hz), 123.4, 43.4, 40.9 (d, J_{C-P} = 66.1 Hz), 30.3. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.7. IR (film) υ_{max} 3060, 2918, 1714, 1177, 1111 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{20}NO_4PH^+$ 394.1203, found 394.1215.

*4-(1-(Diphenylphosphinyl)-3-oxo-3-methylpropyl)benzonitrile (**3l**).* 74% yield, 116 mg; white solid; mp 193.4−194.4 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.87−7.83 (m, 2H), 7.49−7.44 (m, 3H), 7.41−7.5 (m, 6H), 7.29−7.26 (m, 1H), 7.20−7.16 (m, 2H), 4.21 (ddd, J = 9.8, 7.1, 2.5 Hz, 1H), 3.24 (ddd, J = 18.3, 10.2, 5.0 Hz, 1H), 2.90 (ddd, J = 18.3, 10.9, 2.4 Hz, 1H), 1.89 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.5 (d, J_{C-P} = 12.2 Hz), 141.9 (d, J_{C-P} = 5.4 Hz), 132.4 (d, J_{C-P} = 1.8 Hz), 131.9, 131.9 (d, J_{C-P} = 2.4 Hz), 131.1 (d, J_{C-P} = 8.6 Hz), 131.1 (d, J_{C-P} = 29.5 Hz), 130.7 (d, J_{C-P} = 9.1 Hz), 130.5 (d, J_{C-P} = 5.5 Hz), 130.2, 128.8 (d, J_{C-P} = 89.0 Hz), 128.7 (d, J_{C-P} = 89.6 Hz), 118.6, 110.9 (d, J_{C-P} = 2.3 Hz), 43.3, 41.0 (d, J_{C-P} = 66.3 Hz), 30.3. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.8. IR (film) υ_{max} 3056, 2919, 1715, 1184, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{20}NO_2PH^+$ 374.1304, found 374.1297.

*4-(Diphenylphosphinyl)-4-([1,1′-biphenyl]-4-yl)butan-2-one (**3m**).* 98% yield, 166 mg; white solid; mp 215.1−217.2 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.96−7.92 (m, 2H), 7.57−7.47 (m, 7H), 7.40−7.35 (m, 6H), 7.32−7.28 (m, 2H), 7.25−7.22 (m, 2H), 4.31−4.28 (m,1H), 3.36 (ddd, J = 16.6, 10.0, 5.2 Hz, 1H), 2.98 (ddd, J = 17.7, 11.1, 2.1 Hz, 1H), 1.97 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.4 (d, J_{C-P} = 12.6 Hz), 140.5, 139.8 (d, J_{C-P} = 1.6 Hz), 134.9 (d, J_{C-P} = 5.4 Hz), 132.1 (d, J_{C-P} = 1.8 Hz), 131.7 (d, J_{C-P} = 2.6 Hz), 131.6 (d, J_{C-P} = 2.4 Hz), 131.3 (d, J_{C-P} = 8.1 Hz), 131.1 (d, J_{C-P} = 8.6 Hz), 130.9, 130.1 (d, J_{C-P} = 5.4 Hz), 128.9 (d, J_{C-P} = 11.1 Hz), 128.7, 128.2 (d, J_{C-P} = Hz), 127.3, 127.0, 126.9, 43.5, 40.5 (d, J_{C-P} = 67.8 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.9. IR (film) υ_{max} 2924, 1715, 1165, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{28}H_{25}O_2PH^+$ 425.1665, found 425.1662.

*4-(Diphenylphosphinyl)-4-(4-methoxyphenyl)butan-2-one (**3n**).*[30] 99% yield, 150 mg; white solid; mp 205.1−205.8 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.84−7.80 (m, 2H), 7.43−7.36 (m, 5H),

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7.22−7.19 (m, 1H), 7.14−7.10 (m, 4H), 7.59−7.58 (m, 2H), 4.10 (ddd, J = 10.1, 7.5, 2.7 Hz, 1H), 3.56 (s, 3H), 3.18 (ddd, J = 17.5, 10.2, 5.4 Hz, 1H), 2.81 (ddd, J = 17.7, 10.9, 2.6 Hz, 1H), 1.81 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.5 (d, J_{C-P} = 13.0 Hz), 158.6 (d, J_{C-P} = 2.3 Hz), 131.9 (d, J_{C-P} = 2.4 Hz), 131.8, 131.4 (d, J_{C-P} = 2.4 Hz), 131.2 (d, J_{C-P} = 8.5 Hz), 131.1, 131.0 (d, J_{C-P} = 8.6 Hz), 130.7 (d, J_{C-P} = 5.6 Hz), 128.5 (d, J_{C-P} = 91.9 Hz), 128.4 (d, J_{C-P} = 92.5 Hz), 127.6 (d, J_{C-P} = 5.5 Hz), 113.8, 55.1, 43.6, 39.9 (d, J_{C-P} = 69.1 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.8. IR (film) υ_{max} 3054, 2919, 1705, 1186, 1119 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_3PH^+$ 379.1458, found 379.1459.

4-(Diphenylphosphinyl)-4-(naphthalen-1-yl)butan-2-one (**3o**). 95% yield, 151 mg; white solid; mp 212.1−212.5 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 8.12−8.11 (m, 1H), 8.04−7.99 (m, 3H), 7.69−7.64 (m, 2H), 7.56−7.53 (m, 3H), 7.44−7.27 (m, 5H), 7.06−7.04 (m, 1H), 6.94−6.91 (m, 2H), 5.23−5.19 (m, 1H), 3.43−3.37 (m, 1H), 3.23 (ddd, J = 17.6, 12.0, 3.0 Hz, 1H), 1.89 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.3 (d, J_{C-P} = 11.3 Hz), 133.6, 133.0 (d, J_{C-P} = 4.8 Hz), 132.1 (d, J_{C-P} = 1.8 Hz), 132.0 (d, J_{C-P} = 4.5 Hz), 131.8, 131.5 (d, J_{C-P} = 9.0 Hz), 131.2 (d, J_{C-P} = 2.9 Hz), 131.0, 130.6 (d, J_{C-P} = 9.1 Hz), 128.3 (d, J_{C-P} = 147.2 Hz), 128.2 (d, J_{C-P} = 147.9 Hz), 128.6, 127.8 (d, J_{C-P} = 1.5 Hz), 127.1 (d, J_{C-P} = 5.4 Hz), 126.0, 125.4 (d, J_{C-P} = 1.8 Hz), 125.3, 123.0, 44.8, 40.0 (d, J_{C-P} = 68.5 Hz), 30.3. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.9. IR (film) υ_{max} 3054, 2924, 1715, 1187, 1115 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{26}H_{23}O_2PH^+$ 399.1508, found 399.1506.

4-(Diphenylphosphinyl)-4-(furan-2-yl)butan-2-one (**3p**).[29] 73% yield, 99 mg; yellow solid; mp 186.9−189.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.86−7.82 (m, 2H), 7.56−7.52 (m, 3H), 7.50−7.42 (m, 3H), 7.36−7.33 (m, 2H), 7.17−7.16 (m, 1H), 6.17−6.16 (m, 1H), 6.01−6.00 (m, 1H), 4.51 (ddd, J = 10.7, 10.7, 3.1 Hz, 1H), 3.25 (ddd, J = 17.8, 10.5, 5.6 Hz, 1H), 3.00 (ddd, J = 18.0, 10.0, 3.0 Hz, 1H), 2.04 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.8 (d, J_{C-P} = 11.8 Hz), 148.8 (d, J_{C-P} = 6.4 Hz), 141.8 (d, J_{C-P} = 2.7 Hz), 132.1 (d, J_{C-P} = 1.8 Hz), 131.9 (d, J_{C-P} = 2.8 Hz), 131.4 (d, J_{C-P} = 2.6 Hz), 131.3 (d, J_{C-P} = 9.4 Hz), 130.8 (d, J_{C-P} = 1.9 Hz), 130.6 (d, J_{C-P} = 2.4 Hz), 128.5 (d, J_{C-P} = 69.9 Hz), 128.4 (d, J_{C-P} = 69.9 Hz), 110.7 (d, J_{C-P} = 2.7 Hz), 108.7 (d, J_{C-P} = 6.2 Hz), 40.8, 35.5 (d, J_{C-P} = 69.7 Hz), 30.1. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.2. IR (film) υ_{max} 3053, 2920, 1716, 1185, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{20}H_{19}O_3PH^+$ 339.1145, found 339.1145.

4-(Diphenylphosphinyl)-4-(thiophen-2-yl)butan-2-one (**3q**). 99% yield, 140 mg; white solid; mp 163.1−164.5 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.93−7.89 (m, 2H), 7.56−7.49 (m, 5H), 7.40−7.37 (m, 1H), 7.31−7.28 (m, 2H), 7.06−7.05 (m, 1H), 6.93−6.92 (m, 1H), 6.80−6.79 (m, 1H), 4.59 (ddd, J = 10.6, 8.2, 2.8 Hz, 1H), 3.27 (ddd, J = 17.8, 10.3, 5.1 Hz, 1H), 2.93 (ddd, J = 17.8, 10.3, 2.9 Hz, 1H), 1.99 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.8 (d, J_{C-P} = 12.0 Hz), 137.5 (d, J_{C-P} = 6.4 Hz), 132.1 (d, J_{C-P} = 2.7 Hz), 131.7 (d, J_{C-P} = 2.6 Hz), 131.4 (d, J_{C-P} = 6.5 Hz), 131.3 (d, J_{C-P} = 8.4 Hz), 131.1 (d, J_{C-P} = 9.0 Hz), 130.6 (d, J_{C-P} = 12.7 Hz), 128.6 (d, J_{C-P} = 90.8 Hz), 128.5 (d, J_{C-P} = 90.8 Hz), 127.3 (d, J_{C-P} = 6.4 Hz), 126.7 (d, J_{C-P} = 2.5 Hz), 124.9 (d, J_{C-P} = 2.7 Hz), 44.4, 36.3 (d, J_{C-P} = 70.1 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.5. IR (film) υ_{max} 3056, 2913, 1714, 1178, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{20}H_{19}O_2PSH^+$ 355.0916, found 355.0912.

4-(Diphenylphosphinyl)-4-(pyridin-3-yl)butan-2-one (**3r**).[29] 42% yield, 58 mg; pale yellow solid; mp 198.8−199.6 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 8.43 (m, 1H), 8.36 (m, 1H), 7.97−7.93 (m, 2H), 7.79−7.77 (m, 1H), 7.58−7.47 (m, 5H), 7.36−7.32 (m, 1H), 7.28−7.25 (m, 2H), 7.14−7.12 (m, 1H), 4.26 (ddd, J = 10.0, 7.2, 2.7 Hz, 1H), 3.29 (ddd, J = 18.1, 10.0, 5.1 Hz, 1H), 2.99 (ddd, J = 18.1, 11.1, 2.4 Hz, 1H), 1.97 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.5 (d, J_{C-P} = 12.6 Hz), 151.0 (d, J_{C-P} = 6.4 Hz), 148.4 (d, J_{C-P} = 1.9 Hz), 136.6 (d, J_{C-P} = 5.0 Hz), 132.3 (d, J_{C-P} = 2.7 Hz), 132.1 (d, J_{C-P} = 5.3 Hz), 131.8 (d, J_{C-P} = 2.6 Hz), 131.3(d, J_{C-P} = 14.5 Hz), 131.1 (d, J_{C-P} = 8.8 Hz), 130.7 (d, J_{C-P} = 8.8 Hz), 130.5 (d, J_{C-P} = 9.3 Hz), 128.7 (d, J_{C-P} = 79.9 Hz), 128.6 (d, J_{C-P} = 80.5 Hz), 123.2 (d, J_{C-P} = 1.8 Hz), 43.2, 38.1 (d, J_{C-P} = 67.8 Hz), 30.3. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.1. IR (film) υ_{max} 2923, 1730, 1181, 1112 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{21}H_{20}NO_2PH^+$ 350.1304, found 350.1303.

4-(Diphenylphosphinyl)-4-(benzothiophen-3-yl)butan-2-one (**3s**). 92% yield, 149 mg; white solid; mp 192.4−192.7 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.99−7.95 (m, 2H), 7.87−7.85 (m, 1H), 7.71−7.68 (m, 2H), 7.55−7.49 (m, 3H), 7.45−7.41 (m, 2H), 7.27−7.24 (m, 1H), 7.22−7.19 (m, 1H), 7.15−7.11 (m, 1H), 7.07−7.03 (m, 2H), 4.80 (ddd, J = 9.0, 7.4, 3.3 Hz, 1H), 3.29 (ddd, J = 17.9, 8.8, 6.7 Hz, 1H), 3.11 (ddd, J = 17.9, 11.7, 3.3 Hz, 1H), 1.88 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.0 (d, J_{C-P} = 11.0 Hz), 139.6, 138.4 (d, J_{C-P} = 5.2 Hz), 132.1 (d, J_{C-P} = 2.3 Hz), 131.7 (d, J_{C-P} = 26.3 Hz), 131.5, 131.4 (d, J_{C-P} = 8.6 Hz), 131.1 (d, J_{C-P} = 5.5 Hz), 130.9 (d, J_{C-P} = 20.1 Hz), 130.6 (d, J_{C-P} = 9.1 Hz), 128.5 (d, J_{C-P} = 113.5 Hz), 128.4 (d, J_{C-P} = 114.0 Hz), 125.4 (d, J_{C-P} = 6.4 Hz), 124.1 (d, J_{C-P} = 35.9 Hz), 122.5, 121.8, 44.1, 33.3 (d, J_{C-P} = 69.7 Hz), 30.2. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.2. IR (film) υ_{max} 3053, 2919, 1715, 1185, 1115 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{21}O_2PSH^+$ 405.1073, found 405.1075.

4-(Diphenylphosphinyl)nonan-2-one (**3t**).[32] 91% yield, 125 mg; colorless oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.86−7.78 (m, 4H), 7.49−7.41 (m, 6H), 3.20−3.14 (m, 1H), 2.79 (ddd, J = 19.1, 16.2, 3.8 Hz, 1H), 2.68−2.61 (m, 1H), 1.98 (s, 3H), 1.67−1.58 (m, 1H), 1.53−1.43 (m, 1H), 1.29−1.08 (m, 6H), 0.78 (t, J = 6.9 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.9 (d, J_{C-P} = 9.7 Hz), 132.3(d, J_{C-P} = 95.4 Hz), 132.2 (d, J_{C-P} = 96.3 Hz), 131.7 (d, J_{C-P} = 2.7 Hz), 131.6 (d, J_{C-P} = 2.7 Hz), 130.9, 130.9 (d, J_{C-P} = 17.3 Hz), 128.7 (d, J_{C-P} = 10.9 Hz), 128.6 (d, J_{C-P} = 10.9 Hz), 41.8, 31.6 (d, J_{C-P} = 72.4 Hz), 31.5, 30.0, 28.5, 27.2 (d, J_{C-P} = 10.8 Hz), 22.2, 13.8. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 36.7. IR (film) υ_{max} 3057, 2929, 1717, 1190, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{21}H_{27}O_2PH^+$ 343.1821, found 343.1823.

4-(Diphenylphosphinyl)-4-cyclohexylbutan-2-one (**3u**). 85% yield, 121 mg; white solid; mp 212.1−214.3 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.88−7.84 (m, 2H), 7.80−7.77 (m, 2H), 7.51−7.38 (m, 6H), 3.21−3.17 (m, 1H), 2.76 (dd, J = 13.7 Hz, 5.2 Hz, 2H), 2.11−2.09 (m, 2H), 1.92 (s, 3H), 1.75−1.61 (m, 3H), 1.55−1.50 (m, 2H), 1.14−0.94 (m, 5H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 206.4 (d, J_{C-P} = 8.2 Hz), 132.9 (d, J_{C-P} = 94.3 Hz), 132.8 (d, J_{C-P} = 95.6 Hz), 131.5 (d, J_{C-P} = 6.2 Hz), 131.5, 131.0 (d, J_{C-P} = 8.5 Hz), 130.8 (d, J_{C-P} = 8.8 Hz), 128.6 (d, J_{C-P} = 11.0 Hz), 128.5 (d, J_{C-P} = 11.1 Hz), 37.9, 37.7 (d, J_{C-P} = 1.2 Hz), 36.3 (d, J_{C-P} = 71.3 Hz), 32.8 (d, J_{C-P} = 12.6 Hz), 29.8, 29.4 (d, J_{C-P} = 2.4 Hz), 26.6, 26.3, 26.0. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 36.4. IR (film) υ_{max} 3054, 2925, 1710, 1192, 1115 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{27}O_2PH^+$ 355.1821, found 355.1819.

4-(Diphenylphosphinyl)-5-phenylpentan-2-one (**3v**). 77% yield, 112 mg; white solid; mp 178.2−180.3 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.92−7.83 (m, 4H), 7.48−7.40 (m, 6H), 7.19−7.10 (m, 5H), 3.62−3.56 (m, 1H), 2.98. (ddd, J = 13.6, 8.0, 4.7 Hz, 1H), 2.81−2.68 (m, 2H), 2.60 (ddd, J = 18.0, 9.4, 6.4 Hz, 1H), 1.62 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.5 (d, J_{C-P} = 8.1 Hz), 138.2 (d, J_{C-P} = 13.0 Hz), 131.9 (d, J_{C-P} = 2.4 Hz), 131.8 (d, J_{C-P} = 2.4 Hz), 131.7 (d, J_{C-P} = 96.1 Hz), 131.7 (d, J_{C-P} = 96.3 Hz), 131.1 (d, J_{C-P} = 8.8 Hz), 130.9 (d, J_{C-P} = 8.7 Hz), 129.3, 128.8, 128.7 (d, J_{C-P} = 1.8 Hz), 128.6, 128.4, 126.6, 40.7, 34.0, 33.3 (d, J_{C-P} = 71.3 Hz), 29.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 36.1. IR (film) υ_{max} 3054, 2924, 1714, 1179, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_2PH^+$ 363.1508, found 363.1509.

4-(Diphenylphosphinyl)-6-phenylhex-5-en−2-one (**3w**). 86% yield, 129 mg; white solid; mp 202.5−205.7 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.88−7.85 (m, 2H), 7.78−7.75 (m, 2H), 7.53−7.40 (m, 6H), 7.22−7.17 (m, 5H), 6.38 (dd, J = 15.9 Hz, 3.9 Hz, 1H), 6.07 (ddd, J = 15.7, 9.0, 6.5 Hz, 1H), 3.98−3.92 (m, 1H), 2.98 (ddd, J = 17.3, 9.4, 5.1 Hz, 1H), 2.88−2.82 (m, 1H), 2.06−2.05 (m, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.2 (d, J_{C-P} = 12.4 Hz), 136.5 (d, J_{C-P} = 2.6 Hz), 135.2 (d, J_{C-P} = 11.6 Hz), 132.1 (d, J_{C-P} = 1.8 Hz), 131.8 (d, J_{C-P} = 2.6 Hz), 131.4 (d, J_{C-P} = 95.4 Hz), 131.4 (d, J_{C-P} = 8.7 Hz), 131.2 (d, J_{C-P} = 8.6 Hz), 131.2 (d, J_{C-P} = 99.4 Hz),

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128.9 (d, J_{C-P} = 11.2 Hz), 128.5, 128.5 (d, J_{C-P} = 11.5 Hz), 127.7, 126.3, 123.3 (d, J_{C-P} = 7.3 Hz), 41.7, 38.6 (d, J_{C-P} = 69.8 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.7. IR (film) υ_{max} 3053, 2921, 1714, 1182, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{23}O_2PH^+$ 375.1508, found 375.1509.

*4-(Diphenylphosphinyl)-6-phenylhex-5-yn-2-one (**3x**).* 55% yield, 82 mg; yellow oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.99−7.91 (m, 4H), 7.59−7.54 (m, 2H), 7.51−7.48 (m, 4H), 7.26−7.16 (m, 5H), 4.19 (ddd, J = 16.6, 10.4, 2.9 Hz, 1H), 3.16 (ddd, J = 17.5, 9.6, 2.9 Hz, 1H), 2.88 (ddd, J = 17.1, 10.4, 5.2 Hz, 1H), 2.19 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.2 (d, J_{C-P} = 11.6 Hz), 132.4−132.3 (m), 131.9 (d, J_{C-P} = 93.0 Hz), 131.8 (d, J_{C-P} = 93.6 Hz), 130.7, 130.3, 129.5, 128.7 (d, J_{C-P} = 12.1 Hz), 127.4, 128.3, 128.2, 122.6 (d, J_{C-P} = 3.4 Hz), 85.7 (d, J_{C-P} = 8.3 Hz), 85.2 (d, J_{C-P} = 7.7 Hz), 41.9, 30.2, 29.0 (d, J_{C-P} = 70.6 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 31.8. IR (film) υ_{max} 3054, 2922, 1714, 1180, 1118 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{21}O_2PH^+$ 373.1352, found 373.1353.

*4-(Diphenylphosphinyl)-4-methyldecan-2-one (**3y**).* 79% yield, 117 mg; colorless oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 8.02−7.96 (m, 4H), 7.54−7.46 (m, 6H), 3.07 (dd, J = 16.3, 9.5 Hz, 1H), 2.55 (dd, J = 16.3, 10.6 Hz, 1H), 2.09 (s, 3H), 2.81−2.75 (m, 2H), 1.42 (d, J = 16.8 Hz, 3H), 1.38 (m, 1H), 1.19−1.12 (m, 7H), 0.81 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 206.8 (d, J_{C-P} = 12.4 Hz), 132.2, 132.1, 131.6 (d, J_{C-P} = 2.6 Hz), 131.5 (d, J_{C-P} = 2.6 Hz), 131.4 (d, J_{C-P} = 89.2 Hz), 131.0 (d, J_{C-P} = 90.0 Hz), 128.4, 128.3, 44.7, 40.3 (d, J_{C-P} = 67.7 Hz), 34.9, 32.3, 31.4, 29.8, 24.4 (d, J_{C-P} = 4.9 Hz), 22.4, 21.0, 14.0. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 36.5. IR (film) υ_{max} 3055, 2924, 1714, 1177 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{31}O_2PH^+$ 371.2134, found 371.2133.

*4-Methyl-4-(diphenylphosphinyl)-4-phenylbutan-2-one (**3z**).* 99% yield, 142 mg; colorless oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.65−7.61 (m, 2H), 7.54−7.48 (m, 3H), 7.45−7.37 (m, 3H), 7.33−7.29 (m, 2H), 7.23−7.18 (m, 3H), 7.09−7.07 (m, 2H), 3.58 (dd, J = 16.9, 7.9 Hz, 1H), 3.12 (dd, J = 5.4, 16.9 Hz, 1H), 1.90 (s, 3H), 1.79 (d, J = 16.4 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.1 (d, J_{C-P} = 15.4 Hz), 138.4 (d, J_{C-P} = 3.3 Hz), 132.9 (d, J_{C-P} = 7.3 Hz), 132.7 (d, J_{C-P} = 8.0 Hz), 131.9 (d, J_{C-P} = 2.6 Hz), 131.8 (d, J_{C-P} = 2.8 Hz), 129.6 (d, J_{C-P} = 93.1 Hz), 129.2 (d, J_{C-P} = 92.0 Hz), 128.3 (d, J_{C-P} = 4.9 Hz), 128.1 (d, J_{C-P} = 10.9 Hz), 127.9 (d, J_{C-P} = 11.3 Hz), 127.9 (d, J_{C-P} = 2.9 Hz), 127.1 (d, J_{C-P} = 3.6 Hz), 47.3, 44.4 (d, J_{C-P} = 63.1 Hz), 31.8 (d, J_{C-P} = 1.5 Hz), 19.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 35.1. IR (film) υ_{max} 3056, 2928, 1705, 1176, 1107 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_2PH^+$ 363.1508, found 363.1506.

*1-(1-Diphenylphosphinylcyclohexyl)propan-2-one (**3aa**).* 89% yield, 121 mg; white solid; mp 153.3−155.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 8.02−7.98 (m, 4H), 7.53−7.45 (m, 6H), 2.82 (d, J = 19.2 Hz, 2H), 2.24−2.22 (m, 2H), 2.00 (s, 3H), 1.64−1.46 (m, 7H), 1.08−1.00 (m, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 207.6 (d, J_{C-P} = 1.6 Hz), 132.3 (d, J_{C-P} = 8.1 Hz), 131.6 (d, J_{C-P} = 2.3 Hz), 130.9, 128.2 (d, J_{C-P} = 10.9 Hz), 43.1, 42.6, 41.2, 32.5, 29.7, 26.5, 25.2, 20.8 (d, J_{C-P} = 10.8 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 35.5. IR (film) υ_{max} 3056, 2925, 1722, 1179, 1109 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{21}H_{25}O_2PH^+$ 341.1665, found 341.1665.

*3-Methyl-4-(diphenylphosphinyl)-4-phenylbutan-2-one (**3ab**).* 87% yield, 126 mg; white solid; mp 175.9−177.2 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.92−7.89 (m, 2H), 7.55−7.45 (m, 5H), 7.35−7.33 (m, 2H), 7.28−7.26 (m, 1H), 7.22−7.19 (m, 2H), 7.16−7.09 (m, 3H), 4.36 (dd, J = 7.8, 7.8 Hz, 1H), 3.46−3.38 (m, 1H), 1.92 (s, 3H), 1.02 (d, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 210.6 (d, J_{C-P} = 5.5 Hz), 134.4 (d, J_{C-P} = 4.6 Hz), 132.4 (d, J_{C-P} = 94.5 Hz), 132.1 (d, J_{C-P} = 99.1 Hz), 131.8 (d, J_{C-P} = 2.6 Hz), 131.6 (d, J_{C-P} = 8.8 Hz), 131.1 (d, J_{C-P} = 4.5 Hz), 131.1 (d, J_{C-P} = 5.9 Hz), 131.0, 128.5 (d, J_{C-P} = 11.5 Hz), 128.2, 128.0 (d, J_{C-P} = 11.7 Hz), 127.1(d, J_{C-P} = 1.5 Hz), 46.3 (d, J_{C-P} = 67.7 Hz), 45.6, 28.8, 16.0 (d, J_{C-P} = 6.7 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.4. IR (film) υ_{max} 3054, 2924, 1712, 1170, 1115 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{23}H_{23}O_2PH^+$ 363.1508, found 363.1509.

*4-(Diphenylphosphinyl)-1,1,4-triphenylbutan-2-one (**3ac**).* 56% yield, 112 mg. white solid; mp 203.8−206.3 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.90−7.87 (m, 2H), 7.57−7.49 (m, 3H), 7.40−7.37 (m, 2H), 7.31−7.29 (m, 1H), 7.21−7.08 (m, 13H), 6.96−6.94 (m, 2H), 6.75−6.73 (m, 2H), 4.85 (s, 1H), 4.25−4.22 (m, 1H), 3.46 (ddd, J = 17.2, 10.7, 2.2 Hz, 1H), 2.99−2.94 (m, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.3 (d, J_{C-P} = 13.2 Hz), 137.7, 136.9, 135.2, 132.0, 131.4, 131.3 (d, J_{C-P} = 8.2 Hz), 130.9 (d, J_{C-P} = 9.2 Hz), 129.9 (d, J_{C-P} = 5.4 Hz), 129.0 (d, J_{C-P} = 11.2 Hz), 128.9 (d, J_{C-P} = 7.7 Hz), 128.7, 128.5, 128.3, 128.1 (d, J_{C-P} = 11.5 Hz), 127.2, 127.1, 64.6, 42.7, 41.3 (d, J_{C-P} = 68.2 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.5. IR (film) υ_{max} 3054, 22921, 1715, 1182, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{34}H_{29}O_2PH^+$ 501.1978, found 501.1978.

*4-(Diphenylphosphinyl)-1,4-diphenylbutan-2-one (**3ad**).* 25% yield, 42 mg; white solid; mp 251.6−253.2 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.83−7.79 (m, 2H), 7.48−7.40 (m, 3H), 7.35−7.31 (m, 2H), 7.25−7.22 (m, 1H), 7.15−7.09 (m, 7H), 7.03−7.02 (m, 3H), 6.80−6.79 (m, 2H), 4.12 (ddd, J = 10.1, 7.3, 2.8 Hz, 1H), 3.38 (dd, J = 22.9, 15.3 Hz, 2H), 3.27 (ddd, J = 17.6, 10.2, 5.5 Hz, 1H), 2.85 (ddd, J = 17.7, 11.0, 2.9 Hz, 1H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.1 (d, J_{C-P} = 12.7 Hz), 135.5 (d, J_{C-P} = 5.5 Hz), 133.1, 132.0 (d, J_{C-P} = 2.7 Hz), 131.8 (d, J_{C-P} = 24.5 Hz), 131.4 (d, J_{C-P} = 2.4 Hz), 131.3 (d, J_{C-P} = 8.7 Hz), 131.1, 131.0 (d, J_{C-P} = 9.0 Hz), 129.7 (d, J_{C-P} = 5.7 Hz), 129.3, 128.7, 128.5 (d, J_{C-P} = 101.7 Hz), 128.4 (d, J_{C-P} = 102.4 Hz), 128.3, 127.1 (d, J_{C-P} = 1.9 Hz), 127.0, 50.5, 41.9, 41.0 (d, J_{C-P} = 68.4 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.6. IR (film) υ_{max} 3063, 2926, 1714, 1190, 1117 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{28}H_{25}O_2PH^+$ 425.1665, found 425.1658.

*7-(Diphenylphosphinyl)dodecan-5-one (**3ae**).* 42% yield, 65 mg; colorless oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.86−7.78 (m, 4H), 7.51−7.41 (m, 6H), 3.23−3.17 (m, 1H), 2.80−2.73 (m, 1H), 2.64−2.57 (m, 1H), 2.31−2.24 (m, 1H), 2.18−2.12 (m, 1H), 1.67−1.58 (m, 1H), 1.53−1.32 (m, 3H), 1.29−1.09 (m, 8H), 0.82 (t, J = 7.3 Hz, 3H), 0.78 (t, J = 6.9 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 208.5 (d, J_{C-P} = 9.3 Hz), 132.3 (d, J_{C-P} = 95.8 Hz), 132.3 (d, J_{C-P} = 95.0 Hz), 131.6, 131.6 (d, J_{C-P} = 4.5 Hz), 131.0 (d, J_{C-P} = 8.2 Hz), 130.9 (d, J_{C-P} = 8.2 Hz), 128.7, 128.6, 42.7, 40.9, 31.5, 30.9, 28.5 (d, J_{C-P} = 1.4 Hz), 27.2 (d, J_{C-P} = 10.8 Hz), 25.8, 22.2, 22.1, 13.8, 13.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 36.7. IR (film) υ_{max} 2920, 2851, 1714, 1166, 1118 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{33}O_2PH^+$ 385.2291, found 385.2290.

*1-(Diphenylphosphinyl)-1-phenylheptan-3-one (**3af**).*[6] 90% yield, 141 mg; white solid; mp 182.7−185.1 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.88−7.85 (m, 2H), 7.50−7.43 (m, 3H), 7.38−7.35 (m, 2H), 7.25−7.21 (m, 3H), 7.17−7.14 (m, 2H), 7.09−7.03 (m, 3H), 4.17 (ddd, J = 9.8, 7.0, 2.7 Hz, 1H), 3.23 (ddd, J = 17.6, 10.1, 5.5 Hz, 1H), 2.82 (ddd, J = 17.6, 11.3, 2.7 Hz, 1H), 2.17−2.03 (m, 2H), 1.30−1.21 (m, 2H), 1.02−0.95 (m, 2H), 0.67 (t, J = 7.3 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 207.9 (d, J_{C-P} = 12.3 Hz), 135.9 (d, J_{C-P} = 5.5 Hz), 132.0 (d, J_{C-P} = 2.7 Hz), 131.8, 131.4 (d, J_{C-P} = 2.7 Hz), 131.3 (d, J_{C-P} = 9.1 Hz), 131.1 (d, J_{C-P} = 13.2 Hz), 131.0 (d, J_{C-P} = 8.8 Hz), 129.7 (d, J_{C-P} = 6.4 Hz), 128.5 (d, J_{C-P} = 100.3 Hz), 128.4 (d, J_{C-P} = 100.9 Hz), 128.3 (d, J_{C-P} = 1.3 Hz), 127.1 (d, J_{C-P} = 2.4 Hz), 43.3, 42.6, 40.8 (d, J_{C-P} = 68.4 Hz), 25.6, 22.0, 13.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.8. IR (film) υ_{max} 3048, 2918, 1715, 1184, 1118 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{25}H_{27}O_2PH^+$ 391.1821, found 391.1823.

*1-(Diphenylphosphinyl)-1-(4-chlorophenyl)heptan-3-one (**3ag**).* 85% yield, 127 mg; white solid; mp 217.9−219.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.84−7.90 (m, 2H), 7.58−7.51 (m, 3H), 7.48−7.45 (m, 2H), 7.37−7.35 (m, 1H), 7.29−7.24 (m, 4H), 7.14−7.13 (m, 2H), 4.22 (m, 1H), 3.24 (ddd, J = 16.8, 9.9, 3.7 Hz, 1H), 2.90−2.84 (m, 1H), 2.25−2.12 (m, 2H), 1.36−1.29 (m, 2H), 1.11−1.04 (m, 2H), 0.76 (t, J = 7.3 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 207.5 (d, J_{C-P} = 11.9 Hz), 134.7 (d, J_{C-P} = 5.1 Hz), 133.0 (d, J_{C-P} = 1.5 Hz), 132.1, 131.6, 131.3, 131.2, 131.0 (d, J_{C-P} = 4.5 Hz), 130.9, 130.8, 128.6 (d, J_{C-P} = 86.3 Hz), 128.5 (d, J_{C-P} = 87.1 Hz), 128.5, 43.2, 42.6, 40.1 (d, J_{C-P} = 68.0 Hz), 25.5, 22.0, 13.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.2. IR (film) υ_{max} 2922, 1737,

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1183, 1122 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{25}H_{26}ClO_2PH^+$ 425.1432, found 425.1433.

*1-(Diphenylphosphinyl)-1-(4-methylphenyl)heptan-3-one (**3ah**).* 88% yield, 139 mg; white solid; mp 182.6−183.8 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.86−7.82 (m, 2H), 7.43−7.37 (m, 5H), 7.24−7.21 (m, 1H), 7.16−7.10 (m, 4H), 6.89−6.86 (m, 2H), 4.16−4.12 (m, 1H), 3.18 (ddd, J = 17.6, 10.0, 5.5 Hz, 1H), 2.79 (ddd, J = 17.3, 11.2, 2.4 Hz, 1H), 2.12 (s, 3H), 2.15−2.01 (m, 2H), 1.26−1.20 (m, 2H), 1.01−0.94 (m, 2H), 0.65 (t, J = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 207.9 (d, J_{C-P} = 12.2 Hz), 136.6 (d, J_{C-P} = 2.3 Hz), 132.7 (d, J_{C-P} = 5.4 Hz), 132.1 (d, J_{C-P} = 3.9 Hz), 131.9 (d, J_{C-P} = 2.1 Hz), 131.3, 131.2, 131.0 (d, J_{C-P} = 8.7 Hz), 129.6 (d, J_{C-P} = 5.4 Hz), 129.0 (d, J_{C-P} = 1.5 Hz), 128.5 (d, J_{C-P} = 94.9 Hz), 128.4 (d, J_{C-P} = 94.9 Hz), 43.3, 42.7, 40.3 (d, J_{C-P} = 68.8 Hz), 25.5, 22.0, 21.0, 13.7. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.6. IR (film) υ_{max} 2922, 1714, 1184, 1121 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{26}H_{29}O_2PH^+$ 405.1978, found 405.1977.

*1-(Diphenylphosphinyl)-1-(4-(trifluoromethyl)phenyl)heptan-3-one (**3ai**).* 57% yield, 104.5 mg; white solid; mp 225.5−227.2 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.97−7.93 (m, 2H), 7.59−7.52 (m, 3H), 7.49−7.41 (m, 6H), 7.36−7.33 (m, 1H), 7.26−7.24 (m, 2H), 4.35−4.32 (m, 1H), 3.31 (ddd, J = 17.8, 10.1, 5.1 Hz, 1H), 2.93 (ddd, J = 17.8, 11.6, 2.2 Hz, 1H), 2.26−2.13 (m, 2H), 1.36−1.29 (m, 2H), 1.10−1.03 (m, 2H), 0.74 (t, J = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 207.4 (d, J_{C-P} = 11.8 Hz), 140.4 (d, J_{C-P} = 4.2 Hz), 132.2 (d, J_{C-P} = 1.6 Hz), 131.7 (d, J_{C-P} = 2.7 Hz), 131.4 (d, J_{C-P} = 24.9 Hz), 131.2 (d, J_{C-P} = 8.4 Hz), 130.8 (d, J_{C-P} = 9.1 Hz), 130.0 (d, J_{C-P} = 5.4 Hz), 129.2 (d, J_{C-P} = 39.1 Hz), 128.7 (d, J_{C-P} = 90.3 Hz), 128.6 (d, J_{C-P} = 90.9 Hz), 125.2−125.1 (m), 123.0, 43.1, 42.5, 40.7 (d, J_{C-P} = 67.2 Hz), 25.5, 22.0, 13.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 33.2. IR (film) υ_{max} 3054, 2922, 1714, 1175, 1111 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{26}H_{26}F_3O_2PH^+$ 459.1695, found 459.1695.

*4-(Bis(4-methylphenyl)phosphinyl)-4-phenylbutan-2-one (**3aj**).* 99% yield, 149 mg; white solid; mp 170.0−171.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.79−7.75 (m, 2H), 7.36−7.32 (m, 2H), 7.30−7.28 (m, 2H), 7.24−7.22 (m, 2H), 7.13−7.07 (m, 3H), 6.98−6.96 (m, 2H), 4.22−4.18 (m, 1H), 3.31 (ddd, J = 17.5, 10.1, 5.5 Hz, 1H), 2.96 (ddd, J = 17.5, 11.2, 2.1 Hz, 1H), 2.34 (s, 3H), 2.20 (s, 3H), 1.89 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.7 (d, J_{C-P} = 12.6 Hz), 142.4 (d, J_{C-P} = 2.5 Hz), 141.8 (d, J_{C-P} = 2.6 Hz), 136.0 (d, J_{C-P} = 5.4 Hz), 131.3 (d, J_{C-P} = 9.1 Hz), 131.1 (d, J_{C-P} = 9.1 Hz), 129.8 (d, J_{C-P} = 5.6 Hz), 129.2 (d, J_{C-P} = 93.6 Hz), 129.1 (d, J_{C-P} = 94.6 Hz), 128.3, 127.5, 127.4, 127.0 (d, J_{C-P} = 1.7 Hz), 43.5, 41.0 (d, J_{C-P} = 68.6 Hz), 30.5, 21.5, 21.4. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 34.7. IR (film) υ_{max} 3061, 2914, 1715, 1181, 1116 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{25}O_2PH^+$ 377.1665, found 377.1665.

*4-(Bis(4-chlorophenyl)phosphinyl)-4-phenylbutan-2-one (**3ak**).* 95% yield, 159 mg; white solid; mp 171.4−173.3 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.87−7.84 (m, 2H), 7.52−7.51 (m, 2H), 7.37−7.33 (m, 2H), 7.29−7.28 (m, 2H), 7.23−7.15 (m, 5H), 4.23−4.19 (m, 1H), 3.31 (ddd, J = 17.7, 9.8, 5.5 Hz, 1H), 2.93 (ddd, J = 17.7, 11.8, 2.3 Hz, 1H), 1.97 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 204.8 (d, J_{C-P} = 12.4 Hz), 138.9 (d, J_{C-P} = 2.9 Hz), 138.2 (d, J_{C-P} = 3.4 Hz), 135.3 (d, J_{C-P} = 5.4 Hz), 132.6 (d, J_{C-P} = 9.2 Hz), 132.3 (d, J_{C-P} = 9.7 Hz), 130.0 (d, J_{C-P} = 20.9 Hz), 129.6 (d, J_{C-P} = 5.6 Hz), 129.2 (d, J_{C-P} = 15.4 Hz), 129.0 (d, J_{C-P} = 103.4 Hz), 128.9 (d, J_{C-P} = 103.7 Hz), 128.6 (d, J_{C-P} = 1.8 Hz), 127.4 (d, J_{C-P} = 2.4 Hz), 43.4, 40.6 (d, J_{C-P} = 69.4 Hz), 30.6. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 32.8. IR (film) υ_{max} 2922, 1714, 1181, 1086 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{19}Cl_2O_2PH^+$ 417.0572, found 417.0572.

*4-(Bis(4-methoxyphenyl)phosphinyl)-4-phenylbutan-2-one (**3al**).* 94% yield, 154 mg; colorless oil. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.83−7.79 (m, 2H), 7.32−7.26 (m, 4H), 7.18−7.11 (m, 3H), 7.01−7.00 (m, 2H), 6.72−6.71 (m, 2H), 4.15−4.12 (m, 1H), 3.74 (s, 3H), 3.72 (s, 3H), 3.29 (ddd, J = 17.4, 10.0, 5.3 Hz, 1H), 2.96 (ddd, J = 17.5, 11.3, 2.2 Hz, 1H), 1.94 (s, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.6 (d, J_{C-P} = 12.6 Hz), 162.4 (d, J_{C-P} = 2.6 Hz), 161.9 (d, J_{C-P} = 2.7 Hz), 136.2 (d, J_{C-P} = 5.4 Hz), 133.1 (d, J_{C-P} = 9.1 Hz), 132.9 (d, J_{C-P} = 10.2 Hz), 129.7 (d, J_{C-P} = 5.4 Hz), 128.3 (d, J_{C-P} = 1.5 Hz), 127.0 (d, J_{C-P} = 1.8 Hz), 123.0 (d, J_{C-P} = 48.7 Hz), 122.2 (d, J_{C-P} = 43.6 Hz), 114.0 (d, J_{C-P} = 99.9 Hz), 113.9 (d, J_{C-P} = 101.2 Hz), 55.3, 55.1, 43.6, 41.3 (d, J_{C-P} = 68.9 Hz), 30.5. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 34.2. IR (film) υ_{max} 2924, 1714, 1179, 1118 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{24}H_{25}O_4PH^+$ 409.1563, found 409.1564.

*4-(Di-n-butylphosphinyl)-4-phenylbutan-2-one (**3am**).*[33] 67% yield, 83 mg; yellow solid; mp 85.1−87.2 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.32−7.29 (m, 4H), 7.25−7.23 (m, 1H), 3.61 (ddd, J = 13.1, 9.3, 3.8 Hz, 1H), 3.27 (ddd, J = 17.9, 10.0, 3.9 Hz, 1H), 3.21−3.14 (m, 1H), 2.09 (s, 3H), 1.74−1.52 (m, 4H), 1.48−1.29 (m, 6H), 1.26−1.20 (m, 2H), 0.94 (t, J = 7.3 Hz, 3H), 0.79 (t, J = 7.3 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.4 (d, J_{C-P} = 10.1 Hz), 137.1 (d, J_{C-P} = 4.5 Hz), 128.8, 128.7, 127.2 (d, J_{C-P} = 2.1 Hz), 43.0, 39.8 (d, J_{C-P} = 60.2 Hz), 30.3, 26.3 (d, J_{C-P} = 62.8 Hz), 25.4 (d, J_{C-P} = 65.4 Hz), 24.1, 24.0 (d, J_{C-P} = 30.0 Hz), 23.9 (d, J_{C-P} = 3.7 Hz), 23.5 (d, J_{C-P} = 3.8 Hz), 13.5, 13.4. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 50.9. IR (film) υ_{max} 3062, 2928, 1716, 1158, 1093 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{18}H_{29}O_2PH^+$ 309.1978, found 309.1977.

*Ethyl (3-Oxobutyl)(phenyl)phosphinate (**3an**).* 12% yield, 16 mg; white solid; mp 80.2−82.9 °C. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.51−7.47 (m, 2H), 7.18−7.13 (m, 5H), 6.99−6.97 (m, 3H), 3.76−3.67 (m, 3H), 3.12−3.09 (m, 1H), 2.99−2.92 (m, 1H), 1.91 (s, 3H), 1.08 (t, J = 6.9 Hz, 3H) and isomer: δ 7.44−7.29 (m, 5H), 7.24−7.20 (m, 2H), 7.05−7.01 (m, 3H), 4.08−4.00 (m, 1H), 3.89−3.77 (m, 2H), 3.26−3.20 (m, 1H), 3.08−3.04 (m, 1H), 2.01 (s, 3H), 1.22 (t, J = 6.9 Hz, 3H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 203.8 (d, J_{C-P} = 14.0 Hz), 134.7 (d, J_{C-P} = 7.1 Hz), 131.4, 131.2 (d, J_{C-P} = 9.1 Hz), 128.4 (d, J_{C-P} = 5.5 Hz), 128.3 (d, J_{C-P} = 124.4 Hz), 127.3 (d, J_{C-P} = 15.5 Hz), 127.3 (d, J_{C-P} = 5.3 Hz), 126.1 (d, J_{C-P} = 2.7 Hz), 60.2 (m), 41.6 (d, J_{C-P} = 97.1 Hz), 41.6, 29.3, 15.3 (d, J_{C-P} = 6.0 Hz) and isomer 204.2 (d, J_{C-P} = 12.6 Hz), 134.2 (d, J_{C-P} = 3.4 Hz), 131.4, 131.2 (d, J_{C-P} = 9.1 Hz), 128.2 (d, J_{C-P} = 6.1 Hz), 127.9 (d, J_{C-P} = 125.3 Hz), 127.2, 127.2 (d, J_{C-P} = 9.2 Hz), 126.0 (d, J_{C-P} = 2.8 Hz), 60.2 (m), 41.4 (d, J_{C-P} = 2.7 Hz), 41.1 (d, J_{C-P} = 97.7 Hz), 29.5, 15.4 (d, J_{C-P} = 6.1 Hz). $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 41.7 and isomer: δ 43.2. IR (film) υ_{max} 2922, 1713, 1225, 1121 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{18}H_{21}O_3PH^+$ 317.1301, found 317.1302.

*4-(tert-Butyl(phenyl)phosphinyl)-4-phenylbutan-2-one (**3ap**).* 81% yield, 106 mg; white solid. ^1H NMR ($CDCl_3$, 500 MHz): δ 7.45−7.41 (m, 2H), 7.29−7.26 (m, 1H), 7.21−7.17 (m, 4H), 7.05−7.02 (m, 2H), 7.00−6.96 (m, 1H), 4.23 (ddd, J = 9.7, 6.8, 3.2 Hz, 1H), 3.48 (ddd, J = 18.1, 9.7, 3.2 Hz, 1H), 3.17 (ddd, J = 18.2, 9.2, 5.3 Hz, 1H), 2.04 (s, 3H), 1.20 (d, J = 14.3 Hz, 9H) and isomer: δ 7.85−7.82 (m, 2H), 7.64 (d, J = 5.9 Hz, 2H), 7.56−7.49 (m, 3H), 7.36−7.32 (m, 2H), 7.27−7.24 (m, 1H), 4.16−4.14 (m, 1H), 3.17 (ddd, J = 18.2, 9.2, 5.3 Hz, 1H), 2.62−2.56 (m, 1H), 1.81 (s, 3H), 0.84 (d, J = 15.0 Hz, 9H). $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 205.7 (d, J_{C-P} = 10.6 Hz), 138.9 (d, J_{C-P} = 5.1 Hz), 132.3 (d, J_{C-P} = 7.2 Hz), 130.8 (d, J_{C-P} = 2.8 Hz), 129.5 (d, J_{C-P} = 5.7 Hz), 129.4 (d, J_{C-P} = 88.0 Hz), 128.2, 127.3 (d, J_{C-P} = 11.0 Hz), 126.6 (d, J_{C-P} = 1.8 Hz), 46.0, 37.4 (d, J_{C-P} = 58.9 Hz), 34.7 (d, J_{C-P} = 63.7 Hz), 30.7, 25.5 and isomer: δ 205.4 (d, J_{C-P} = 11.9 Hz), 137.6 (d, J_{C-P} = 4.5 Hz), 131.7 (d, J_{C-P} = 6.4 Hz), 131.6, 130.6 (d, J_{C-P} = 82.7 Hz), 130.1 (d, J_{C-P} = 5.0 Hz), 128.7, 128.5 (d, J_{C-P} = 10.3 Hz), 127.4, 46.0, 37.3 (d, J_{C-P} = 60.3 Hz), 33.1 (d, J_{C-P} = 67.0 Hz), 30.6, 24.9. $^{31}P\{^1H\}$ NMR ($CDCl_3$, 202 MHz): δ 49.8 and isomer: δ 51.3. IR (film) υ_{max} 3060, 2973, 1714, 1362, 1157 cm^{-1}. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{20}H_{25}O_2PH^+$ 329.1665, found 329.1662.

*Procedure for the Synthesis of **4**.* To a stirred solution of the **3a** (1 mmol) in 1.7 mL of DMSO was added slowly 0.44 mL (3 mmol) of 48% aqueous HBr. The solution was stirred in an open flask at 55 °C in an oil bath for 4 h. Then the solution was poured onto ice. The solid products were filtered, washed with water, and dried under vacuum. The resulting mixture was purified by silica gel chromatography to afford **4**.

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*3-Bromo-4-(diphenylphosphinyl)-4-phenylbutan-2-one (**4**).* 56% yield, 239 mg; white solid. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.89−7.85 (m, 2H), 7.54−7.52 (m, 1H), 7.50−7.46 (m, 2H), 7.43−7.40 (m, 2H), 7.34−7.31 (m, 1H), 7.23−7.19 (m, 2H), 7.14−7.08 (m, 5H), 5.19 (dd, J = 11.3, 7.4 Hz, 1H), 4.68 (dd, J = 11.3, 10.3 Hz, 1H), 2.29 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 199.6, 135.0 (d, J_{C-P} = 4.6 Hz), 132.4, 132.2 (d, J_{C-P} = 2.7 Hz), 131.6 (d, J_{C-P} = 2.9 Hz), 131.5 (d, J_{C-P} = 7.4 Hz), 130.9, 130.3 (d, J_{C-P} = 5.2 Hz), 130.1, 128.4 (d, J_{C-P} = 87.5 Hz), 128.3 (d, J_{C-P} = 87.8 Hz), 128.2 (d, J_{C-P} = 1.6 Hz), 127.6 (d, J_{C-P} = 2.5 Hz), 52.8 (d, J_{C-P} = 6.1 Hz), 50.0 (d, J_{C-P} = 57.9 Hz), 27.4. ^{31}P{^{1}H} NMR ($CDCl_3$, 202 MHz): δ 30.9. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{22}H_{20}BrO_2PH^+$ 427.0457, found 427.0455.

*Procedure for the Synthesis of **5**.* To a Schlenk tube equipped with a magnetic stir bar was added 1-phenyl-2,3-butadien-1-ol (1 mmol) and TfOH (0.3 mmol) in THF (5 mL) under argon, and the mixture was kept stirring at 100 °C in an oil bath for 24 h. The resulting mixture was directly purified by silica gel chromatography to afford **5**.

*(E)-4-Phenyl-3-buten-2-one (**5**).* 51% yield, 75 mg; yellow liquid. ^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.54−7.52 (m, 2H), 7.50 (d, J = 13.7 Hz, 1H),7.39−7.37 (m, 3H), 6.70 (d, J = 16.4 Hz, 1H), 2.37 (s, 3H). ^{13}C{^{1}H} NMR ($CDCl_3$, 125 MHz): δ 198.4, 143.4, 134.4, 130.5, 129.0, 128.3, 127.1, 27.5. HRMS (ESI-TOF) m/z $[M + H]^+$ calcd for $C_{10}H_{10}OH^+$ 147.0804, found 147.0804.

*Procedure for the Synthesis of **3a** via the Addition of **2a** to **5**.* To a Schlenk tube equipped with a magnetic stir bar was added **4** (0.4 mmol), diphenylphosphine oxide (0.48 mmol), and TfOH (0.12 mmol) in THF (2 mL) under argon, and the mixture was kept stirring at 100 °C in an oil bath for 24 h. The resulting mixture was directly purified by silica gel chromatography using a mixture of petroleum ether and ethyl acetate as eluent to afford **3a**.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c00328.

Crystallographic data for compound **3e** (CIF)

X-ray crystallographic data for compounds **3e** and copies of NMR spectra for all products (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) For selected recent publications, see: (a) Sheng, X. C.; Pyun, H. J.; Chaudhary, K.; Wang, J.; Doerffler, E.; Fleury, M.; McMurtrie, D.; Chen, X.; Delaney, W. E.; Kim, C. U. Discovery of Novel Phosphonate Derivatives as Hepatitis C Virus NS3 Protease Inhibitors. *Bioorg. Med. Chem. Lett.* **2009**, *19*, 3453. (b) Leung, P.-H. Asymmetric Synthesis and Organometallic Chemistry of Functionalized Phosphines Containing Stereogenic Phosphorus Centers. *Acc. Chem. Res.* **2004**, *37*, 169. (c) Queffelec, C.; Petit, M.; Janvier, P.; Knight, D. A.; Bujoli, B. Surface Modification Using Phosphonic Acids and Esters. *Chem. Rev.* **2012**, *112*, 3777. (d) Chen, X.; Kopecky, D. J.; Mihalic, J.; Jeffries, S.; Min, X.; Heath, J.; Deignan, J.; Lai, S.; Fu, Z.; Guimaraes, C.; Shen, S.; Li, S.; Johnstone, S.; Thibault, S.; Xu, H.; Cardozo, M.; Shen, W.; Walker, N.; Kayser, F.; Wang, Z. Structure-Guided Design, Synthesis, and Evaluation of Guanine-Derived Inhibitors of the eIF4E mRNA-Cap Interaction. *J. Med. Chem.* **2012**, *55*, 3837. (e) Xie, P.; Fan, J.; Liu, Y.; Wo, X.; Fu, W.; Loh, T.-P. Bronsted Acid/Organic Photoredox Cooperative Catalysis: Easy Access to Tri- and Tetrasubstituted Alkenylphosphorus Compounds from Alcohols and P-H Species. *Org. Lett.* **2018**, *20*, 3341. (f) Wo, X.; Xie, P.; Fu, W.; Gao, C.; Liu, Y.; Sun, Z.; Loh, T.-P. Barium-catalyzed C-OH/P-H dehydrative cross-coupling for C-P bond construction. *Chem. Commun.* **2018**, *54*, 11132. (g) Chen, L.; Zou, Y.-X.; Liu, X.-Y.; Gou, X.-J. Dehydrative Cross-Coupling and Related Reactions between Alcohols (C-OH) and P(O)-H Compounds for C-P Bond Formation. *Adv. Synth. Catal.* **2019**, *361*, 3490.

(2) For representative publications, see: (a) Mori, I.; Iwasaki, G.; Scheidegger, A.; Koizumi, S.; Hayakawa, K.; Mano, J. Preparation of Triazolyalkylphosphonates as Herbicides and Fungicides. PCT Int. Appl. WO 92-JP485 920417; *Chem. Abstr.* **1993**, *118*, 124547. (b) Isomura, Y.; Sakamoto, S.; Abe, T. (Jpn. Kokai Tokkyo Koho) Preparation of 3-Oxopropylidene-1,1-diphosphonates as Inflammation Inhibitors, Analgesics, and Antipyretics. JP 63295595 A 19881201, 1988; *Chem. Abstr.*, **1989**, *111*, 174388. (c) Chakravarty, P. K.; Greenlee, W. J.; Parsons, W. H.; Patchett, A. A.; Combs, P.; Roth, A.; Busch, R. D.; Mellin, T. N. (3-Amino-2-oxoalkyl)-Phosphonic Acids and Their Analogs as Novel Inhibitors of D-alanine: D-alanine Ligase. *J. Med. Chem.* **1989**, *32*, 1886.

(3) For representative publications, see: (a) Enders, D.; Saint-Dizier, A.; Lannou, M. I.; Lenzen, A. The Phospha-Michael Addition in Organic Synthesis. *Eur. J. Org. Chem.* **2006**, *2006*, 29. (b) Strappaveccia, G.; Bianchi, L.; Ziarelli, S.; Santoro, S.; Lanari, D.; Pizzo, F.; Vaccaro, L. PS-BEMP as a Basic Catalyst for the Phospha-Michael Addition to Electron-poor Alkenes. *Org. Biomol. Chem.* **2016**, *14*, 3521. (c) Zhao, D.; Mao, L.; Yang, D.; Wang, R. Zinc-Mediated Asymmetric Additions of Dialkylphosphine Oxides to α,β-Unsatu-

rated Ketones and N-Sulfinylimines. *J. Org. Chem.* **2010**, *75*, 6756. (d) Keglevich, G.; Sipos, M.; Takacs, D.; Greiner, I. A study on the Michael Addition of Dialkylphosphites to Methyl Vinyl Ketone. *Heteroat. Chem.* **2007**, *18*, 226.
(4) Hans, M.; Delaude, L.; Rodriguez, J.; Coquerel, Y. N-Heterocyclic Carbene Catalyzed Carba-, Sulfa-, and Phospha-Michael Additions with NHC·CO_2 Adducts as Precatalysts. *J. Org. Chem.* **2014**, *79*, 2758.
(5) Chu, X.-Q.; Zi, Y.; Meng, H.; Xu, X.-P.; Ji, S.-J. Radical Phosphinylation of α,α-Diaryl Allylic Alcohols with Concomitant 1,2-Aryl Migration. *Chem. Commun.* **2014**, *50*, 7642.
(6) Shan, C.; Chen, F.; Pan, J.; Gao, Y.; Xu, P.; Zhao, Y. $Zn(OTf)_2$-Catalyzed Phosphinylation of Propargylic Alcohols: Access to γ-Ketophosphine Oxides. *J. Org. Chem.* **2017**, *82*, 11659.
(7) Yin, Y.; Weng, W.-Z.; Sun, J.-G.; Zhang, B. Eosin Y-Catalyzed, Visible-light-promoted Carbophosphinylation of Allylic Alcohols via a Radical Neophyl Rearrangement. *Org. Biomol. Chem.* **2018**, *16*, 2356.
(8) (a) Trost, B. M.; Tracy, J. S.; Yusoontorn, T. Vanadium-Catalyzed Coupling of Allenols with Electrophilic Halide Sources for the Formation of α-Halo-α',β'-unsaturated Ketones. *Org. Lett.* **2019**, *21*, 1207. (b) He, Y.; Zheng, Z.; Liu, Q.; Song, G.; Sun, N.; Chai, X. Tunable Synthesis of 2-Ene-1,4-diones, 4-Hydroxycyclopent-2-en-1-ones, and 2-(Furan-3-yl)acetamides via Palladium-Catalyzed Cascade Reactions of Allenols. *J. Org. Chem.* **2018**, *83*, 12514. (c) Li, J.; Kong, W.; Yu, Y.; Fu, C.; Ma, S. Studies on Electrophilic Reaction of Tertiary 2,3-Allenols with NBS in H_2O or Aqueous MeCN: An Efficient Selective Synthesis of 2-Bromoallylic Ketones, 1,2-Allenyl Ketones, or 3-Bromo-2,5-dihydrofurans. *J. Org. Chem.* **2009**, *74*, 8733. (d) Sabbasani, V. R.; Mamidipalli, P.; Lu, H.; Xia, Y.; Lee, D. Subtle Electronic Effects in Metal-Free Rearrangement of Allenic Alcohols. *Org. Lett.* **2013**, *15*, 1552.
(9) Guðmundsson, A.; Gustafson, K. P. J.; Mai, B. K.; Yang, B.; Himo, F.; Bäckvall, J.-E. Efficient Formation of 2,3-Dihydrofurans via Iron-Catalyzed Cycloisomerization of α-Allenols. *ACS Catal.* **2018**, *8*, 12.
(10) Zhou, Z.; Liu, J.; Lu, X. Regiocontrolled Coupling of Aromatic and Vinylic Amides with α-Allenols To Form γ-Lactams via Rhodium(III)-Catalyzed C-H Activation. *Org. Lett.* **2016**, *18*, 5668.
(11) Sai, M.; Matsubara, S. Copper-Catalyzed Direct and Stereoselective Synthesis of Conjugated Enynes from α-Allenols. *Adv. Synth. Catal.* **2019**, *361*, 39.
(12) Alcaide, B.; Almendros, P.; Cembellin, S.; Martínez del Campo, T.; Palop, G. Allenols versus Allenones: Rhodium-Catalyzed Regiodivergent and Tunable Allene Reactivity with Triazoles. *Chem. - Eur. J.* **2017**, *23*, 13754.
(13) Guo, B.; Huang, X.; Fu, C.; Ma, S. Carbazoles from the [4C +2C] Reaction of 2,3-Allenols with Indoles. *Chem. - Eur. J.* **2016**, *22*, 18343.
(14) Li, S.; Miao, B.; Yuan, W.; Ma, S. Carbometalation-Carboxylation of 2,3-Allenols with Carbon Dioxide: A Dramatic Effect of Halide Anion. *Org. Lett.* **2013**, *15*, 977.
(15) (a) Gao, Y.; Wang, G.; Chen, L.; Xu, P.; Zhao, Y.; Zhou, Y.; Han, L.-B. Copper-Catalyzed Aerobic Oxidative Coupling of Terminal Alkynes with H-Phosphonates Leading to Alkynylphosphonates. *J. Am. Chem. Soc.* **2009**, *131*, 7956. (b) Ma, D.; Pan, J.; Yin, L.; Xu, P.; Gao, Y.; Yin, Y.; Zhao, Y. Copper-Catalyzed Direct Oxidative C-H Functionalization of Unactivated Cycloalkanes into Cycloalkyl Benzo[*b*]phosphole Oxides. *Org. Lett.* **2018**, *20*, 3455. (c) Pan, J.; Zhao, R.; Guo, J.; Ma, D.; Xia, Y.; Gao, Y.; Xu, P.; Zhao, Y. Three-component 3-(Phosphoryl)methylindole Synthesis from Indoles, H-phosphine Oxides and Carbonyl Compounds under Metal-free Conditions. *Green Chem.* **2019**, *21*, 792. (d) Chen, F.; Xia, Y.; Lin, R.; Gao, Y.; Xu, P.; Zhao, Y. Copper-Catalyzed Direct Twofold C-P Cross-Coupling of Unprotected Propargylic 1,4-Diols: Access to 2,3-Bis(diarylphosphinyl)-1,3-butadienes. *Org. Lett.* **2019**, *21*, 579.
(16) (a) Dolomanov, O. L.; Bourhis, L. J.; Gildea, R. J.; Howard, J. A. K.; Puschmann, H. OLEX2: A Complete Structure Solution, Refinement and Analysis Program. *J. Appl. Crystallogr.* **2009**, *42*, 339. (b) The supplementary crystallographic data of **3e** (CCDC 1982767) can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.
(17) (a) Li, Y.; Lu, L.-Q.; Das, S.; Pisiewicz, S.; Junge, K.; Beller, M. Highly Chemoselective Metal-Free Reduction of Phosphine Oxides to Phosphines. *J. Am. Chem. Soc.* **2012**, *134*, 18325. (b) Wahbi, A.; Touil, S. Synthesis of New Phosphonoamide and Phosphonocaprolactam Derivatives via the Diethyl Chlorophosphate-Promoted Beckmann Rearrangement of γ-Phosphonyloximes. *Heteroat. Chem.* **2015**, *26*, 397.
(18) (a) Alcaide, B.; Almendros, P.; Cembellin, P.; Martínez del Campo, T. Acid-Catalyzed Synthesis of α,β-Disubstituted Conjugated Enones by a Meyer-Schuster-Type Rearrangement in Allenols. *Adv. Synth. Catal.* **2015**, *357*, 1070. (b) Kaden, S.; Reissig, H. U.; Brüdgam, H.; Hartl, H. Additions of Lithiated Alkoxyallenes to Phthalimide: A New Synthesis of Pyrroloisoindolones and an Unusual Olefination Reaction. *Synthesis* **2006**, 1351.
(19) Grayson, M.; Farley, C. E.; Streuli, C. A. Secondary Phosphine Oxides. The Effect of Structure on Acid Strength and Rates of Cleavage of Disulfides. *Tetrahedron* **1967**, *23*, 1065.
(20) Du, J.-Y.; Ma, Y.-H.; Yuan, R.-Q.; Xin, N.; Nie, S.-Z.; Ma, C.-L.; Li, C.-Z.; Zhao, C.-Q. Metal-Free One-Pot Synthesis of 3-Phosphinoylbenzofurans via Phospha-Michael Addition/Cyclization of H-Phosphine Oxides and in Situ Generated ortho-Quinone Methides. *Org. Lett.* **2018**, *20*, 477.
(21) (a) Luo, H.; Ma, S. CuI-Catalyzed Synthesis of Functionalized Terminal Allenes from 1-Alkynes. *Eur. J. Org. Chem.* **2013**, *2013*, 3041. (b) Petrone, D. A.; Isomura, M.; Franzoni, I.; Rössler, S. L.; Carreira, E. M. Allenylic Carbonates in Enantioselective Iridium-Catalyzed Alkylations. *J. Am. Chem. Soc.* **2018**, *140*, 4697.
(22) (a) Ye, F.; Ma, X.; Xiao, Q.; Li, H.; Zhang, Y.; Wang, J. C(sp)-C(sp^3) Bond Formation through Cu-Catalyzed Cross-Coupling of N-Tosylhydrazones and Trialkylsilylethyne. *J. Am. Chem. Soc.* **2012**, *134*, 5742. (b) Huang, X.; Bugarin, A. Direct Synthesis of α-Allenols from TMS-Protected Alkynes and Aldehydes Mediated by Tetrabutylammonium Fluoride. *Chem. - Eur. J.* **2016**, *22*, 12696.
(23) Zhang, J.; Ye, J.; Ma, S. Harmony of CdI_2 with CuBr for the One-pot Synthesis of Optically Active α-Allenols. *Org. Biomol. Chem.* **2015**, *13*, 4080.
(24) Kuang, J.; Luo, H.; Ma, S. Copper (I) Iodide-Catalyzed One-Step Preparation of Functionalized Allenes from Terminal Alkynes: Amine Effect. *Adv. Synth. Catal.* **2012**, *354*, 933.
(25) Ke, J.; Tang, Y.; Yi, H.; Li, Y.; Cheng, Y.; Liu, C.; Lei, A. Copper-Catalyzed Radical/Radical C_{sp3}-H/P-H Cross-Coupling: α-Phosphorylation of Aryl Ketone *O*-Acetyloximes. *Angew. Chem., Int. Ed.* **2015**, *54*, 6604.
(26) Nunez-Villanueva, D.; Hunter, C. A. Homochiral Oligomers with Highly Flexible Backbones form Stable H-bonded Duplexes. *Chem. Sci.* **2017**, *8*, 206.
(27) Han, Z. S.; Wu, H.; Xu, Y.; Zhang, Y.; Qu, B.; Li, Z.; Caldwell, D. R.; Fandrick, K. R.; Zhang, L.; Roschangar, F.; Song, J. J.; Senanayake, C. H. General and Stereoselective Method for the Synthesis of Sterically Congested and Structurally Diverse P-Stereogenic Secondary Phosphine Oxides. *Org. Lett.* **2017**, *19*, 1796.
(28) Moglie, Y.; González-Soria, M. J.; Martín-García, I.; Radivoy, G.; Alonso, F. Catalyst- and Solvent-free Hydrophosphination and Multicomponent Hydrothiophosphination of Alkenes and Alkynes. *Green Chem.* **2016**, *18*, 4896.
(29) Goryunov, E. I.; Bodrin, G. V.; Goryunova, I. B.; Nelyubina, Y. V.; Petrovskii, P. V.; Strelkova, T. V.; Peregudov, A. S.; Matveeva, A. G.; Pasechnik, M. P.; Matveev, S. V.; Nifant'ev, E. E. β-Diphenylphosphorylated Alkanones and Related Compounds: Synthesis and Structure. *Russ. Chem. Bull.* **2013**, *62*, 780.
(30) Wen, S.; Li, P.; Wu, H.; Yu, F.; Liang, X.; Ye, J. Enantioselective Organocatalytic Phospha-Michael Reaction of α,β-Unsaturated Ketones. *Chem. Commun.* **2010**, *46*, 4806.
(31) Hao, X.-Q.; Huang, J.-J.; Wang, T.; Lv, J.; Gong, J.-F.; Song, M.-P. PCN Pincer Palladium (II) Complex Catalyzed Enantioselective Hydrophosphination of Enones: Synthesis of Pyridine-

https://dx.doi.org/10.1021/acs.joc.0c00328
J. Org. Chem. 2020, 85, 8185−8195

Functionalized Chiral Phosphine Oxides as $NC_{sp3}O$ Pincer Preligands. *J. Org. Chem.* **2014**, *79*, 9512.
(32) Lemport, P. S.; Bodrin, G. V.; Belyakov, A. I.; Petrovskii, P. V.; Vologzhanina, A. V.; Nifant'ev, E. E. Friedlander Reaction in the Synthesis of 2-(Phosphoryl)alkyl-substituted 1,6-Naphthyridines. *Mendeleev Commun.* **2009**, *19*, 303.
(33) Pudovik, A. N.; Sobanov, A. A.; Bakhtiyarova, I. V.; Zimin, M. G. Reactions of Benzalacetophenone, Dibenzal-and Benzalacetone with Phosphonous and Phosphinous Acids and Sodium Diethylphosphide. *Zh. Obshch. Khim.* **1983**, *53*, 2456.

https://dx.doi.org/10.1021/acs.joc.0c00328
J. Org. Chem. 2020, 85, 8185−8195